

Mechatronics 3DX4

Slides 2: Modelling in the Frequency Domain

Dr. O. Onaizah

Department of Computing and Software
McMaster University

Material based on lecture notes by P. Taylor and M. Lawford, *Control Systems Engineering (4th ed.)* by N. Nise, and *Calculus with Analytic Geometry (3rd ed.)* by Larson et al.

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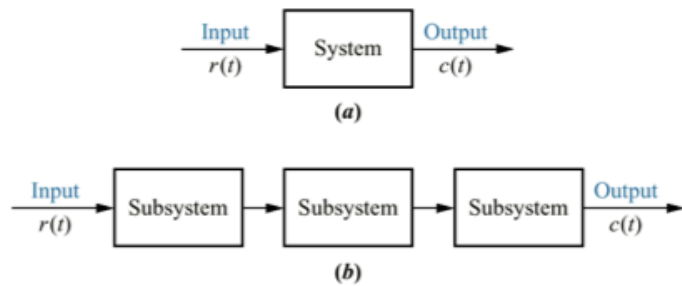
Block Diagram Representation of a System

- ▶ Differential equations can be used to represent relationship between input and output of a system.
- ▶ **Problem:** system parameters, and input ($r(t)$) and output ($c(t)$) appear throughout equation.
- ▶ Prefer to represent system as in Fig. 2.1(a) below where input and output are separate.

Figure 2.1

Want to be able to represent system as series of cascading subsystems, which can easily be combined together.

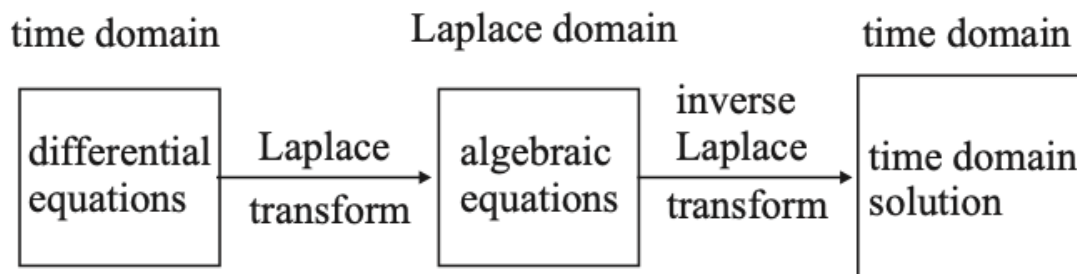
This can not be achieved with differential equations.



Note: The input, $r(t)$, stands for *reference input*.
The output, $c(t)$, stands for *controlled variable*.

Laplace Transform Review

- ▶ Helps us understand the dynamic behaviour of processes.
- ▶ Essential for: stability analysis, block diagrams, and controller design.
- ▶ Converts differential equations (time domain) into algebraic equations (s-domain).



Laplace Transform Definition

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$$

- ▶ $\mathcal{L}$ is the Laplace transform operator.
- ▶ $s = \sigma + j\omega$ is the Laplace transform variable and is a complex number.
- ▶ A sufficient condition for existence of Laplace transform is function is *piecewise continuous* in time over interval of integration.
- ▶ $F(s)$ contains no information about $f(t)$ for time $t < 0$.
- ▶ When doing the inverse Laplace transform, we can represent this by multiplying $f(t)$ by $u(t)$, the unit step function.

Common Laplace Transforms

Table 2.1

- ▶ Table shows the Laplace transform for several common signals.
- ▶ Rather than evaluating Laplace transform using its definition, common to use table and apply various theorems.

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Important Properties of Laplace Transform

Linearity: $\mathcal{L}\{k_1 f_1(t) \pm k_2 f_2(t)\} = k_1 F_1(s) \pm k_2 F_2(s)$

Differentiation:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0-)$$

$$\mathcal{L}\left\{\frac{d^2 f(t)}{dt^2}\right\} = s^2 F(s) - sf(0-) - f'(0-)$$

Frequency Shifting: $\mathcal{L}\{e^{-at} f(t)\} = F(s + a)$

Laplace Transform Theorems

Table 2.2

Commonly
used theorems.

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2 f}{dt^2}\right] = s^2 F(s) - sf(0-) - \dot{f}(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^n f}{dt^n}\right] = s^n F(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0-)$	Differentiation theorem

10.	$\mathcal{L}\left[\int_{0-}^{\cdot} f(\tau) d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

¹ For this theorem to yield correct finite results, all roots of the denominator of $F(s)$ must have negative real parts and no more than one can be at the origin.

² For this theorem to be valid, $f(t)$ must be continuous or have a step discontinuity at $t = 0$ (i.e., no impulses or their derivatives at $t = 0$).

对于FVT, $sF(s)$ 必须是stable的, 所以 $F(s)$ 必须最多有一个 $s=0$ 的pole, 其他pole都在s plain的左半侧。

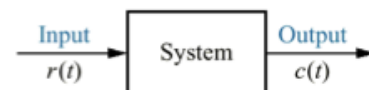
对于IVT, $f(t)$ 必须是连续的。

The Transfer Function

- ▶ We are now ready to represent system in the form of the diagram below.
- ▶ We will separate the system into the following distinct parts: system input, output, and transfer function.
- ▶ In general, an n^{th} order *linear, time-invariant (LTI)* differential equation is of the form:

$$\begin{aligned}
 a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \dots + a_0 c(t) \\
 = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \dots + b_0 r(t)
 \end{aligned}$$

- ▶ $c(t)$ is output, $r(t)$ is input, and differential equation represents system's behavior.



The Transfer Function - II

- ▶ Taking Laplace transform of both sides, gives algebraic equation.

$$a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \dots + a_0 C(s) + \text{init terms for } c(t) \\ = b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \dots + b_0 R(s) + \text{init terms for } r(t)$$

- If we assume **initial conditions are zero**, we can reduce this to:

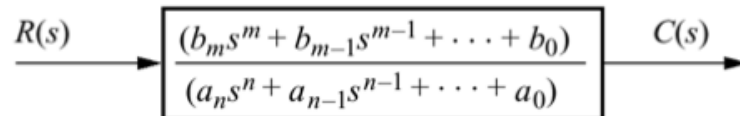
$$(a_n s^n + a_{n-1} s^{n-1} + \dots + a_0) C(s) \\ = (b_m s^m + b_{m-1} s^{m-1} + \dots + b_0) R(s)$$



$$\frac{C(s)}{R(s)} = G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

- $G(s)$ is called the **transfer function**.

- Can find output by
 $C(s) = R(s)G(s)$.



Transfer Function Example

- A given system can be represented by the following differential equation.

$$\tau_p \frac{dc(t)}{dt} + c(t) = k_p r(t)$$

- We will now derive a transfer function for the system.
► Taking the Laplace transform of both sides gives:

$$\mathcal{L}\left\{\tau_p \frac{dc(t)}{dt} + c(t)\right\} = \mathcal{L}\{k_p r(t)\}$$

- Using linearity, we get:

$$\tau_p \mathcal{L}\left\{\frac{dc(t)}{dt}\right\} + \mathcal{L}\{c(t)\} = k_p \mathcal{L}\{r(t)\} \quad (1)$$

- Using differentiation theorem, we get:

$$\mathcal{L}\left\{\frac{dc(t)}{dt}\right\} = sC(s) - c(0-)$$

Transfer Function Example - II

- ▶ When applying Laplace transform to dynamic systems, we typically assume that all inputs are zero at $t < 0$, thus output responses will also be zero until $t = 0$.
- ▶ We can thus rewrite (1) as follows:

$$\tau_p \mathcal{L}\left\{\frac{dc(t)}{dt}\right\} + \mathcal{L}\{c(t)\} = k_p \mathcal{L}\{r(t)\}$$

$$\tau_p s C(s) - c(0-) + C(s) = k_p R(s)$$

$$\tau_p s C(s) + C(s) = k_p R(s)$$

$$\Rightarrow (\tau_p s + 1)C(s) = k_p R(s)$$

- ▶ Thus:

$$G(s) = \frac{C(s)}{R(s)} = \frac{k_p}{\tau_p s + 1} = \frac{\frac{k_p}{\tau_p}}{s + \frac{1}{\tau_p}}$$

Inverse Laplace Transform

- ▶ Once we have solved for the output as a function of the input in the s -domain, we typically want to convert this back into the time domain.
- ▶ We use the inverse Laplace transform:

$$\mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \lim_{\omega \rightarrow \infty} \int_{\sigma - j\omega}^{\sigma + j\omega} F(s) e^{st} ds$$

- ▶ Normally, people look up $F(s)$ in tables, and make use of the theorems.

Partial Fraction Expansion

- ▶ The Laplace transform model of a system will typically be of form:

$$F(s) = \frac{N(s)}{D(s)}$$

- ▶ where $N(s)$ is an m^{th} order polynomial in s and $D(s)$ is an n^{th} order polynomial in s .
- ▶ For example, for $F(s)$ below $m = 2$ and $n = 5$:

$$F(s) = \frac{s^2 + 2s - 3}{s^5 + s^4 - s - 1}$$

- ▶ We know from algebra that all polynomials that have real coefficients can be factored into linear and irreducible quadratic factors.
- ▶ For example: $s^5 + s^4 - s - 1 = (s - 1)(s + 1)^2(s^2 + 1)$

Partial Fraction Expansion - II

- ▶ Using **partial fraction decomposition** and this factorization, we can write the transfer function as follows:

$$\frac{s^2 + 2s - 3}{s^5 + s^4 - s - 1}$$

$$\begin{aligned}
 F'(s) &= \frac{1}{(s-1)(s+1)^2(s^2+1)} \\
 &= \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{Ds+E}{s^2+1}
 \end{aligned}$$

- ▶ To complete the decomposition for this example, we would then determine values for A , B , C , D , and E that satisfy the equation.
- ▶ We can then apply the linearity and the frequency shifting properties to easily find the inverse Laplace transform.

Steps for Decomposing $\frac{N(s)}{D(s)}$ into Partial Fractions

1. **Divide if Improper:** If $\frac{N(s)}{D(s)}$ improper fraction (ie. degree of $N(s)$ is $\geq$ degree of $D(s)$) then divide $N(s)$ by $D(s)$ to get

$$\frac{N(s)}{D(s)} = (\text{a polynomial}) + \frac{N_1(s)}{D(s)}$$

where $\frac{N_1(s)}{D(s)}$ is not an improper fraction. We would then apply the following steps to $\frac{N_1(s)}{D(s)}$.

2. **Factor Denominator:** Factor denominator into factors of form:

$$(ps + q)^m \quad \text{and} \quad (as^2 + bs + c)^n$$

where $as^2 + bs + c$ can not be further reduced.

Steps for Decomposing $\frac{N(s)}{D(s)}$ - II

3. Linear Factors: For factor $(ps + q)^m$, include the sum below:

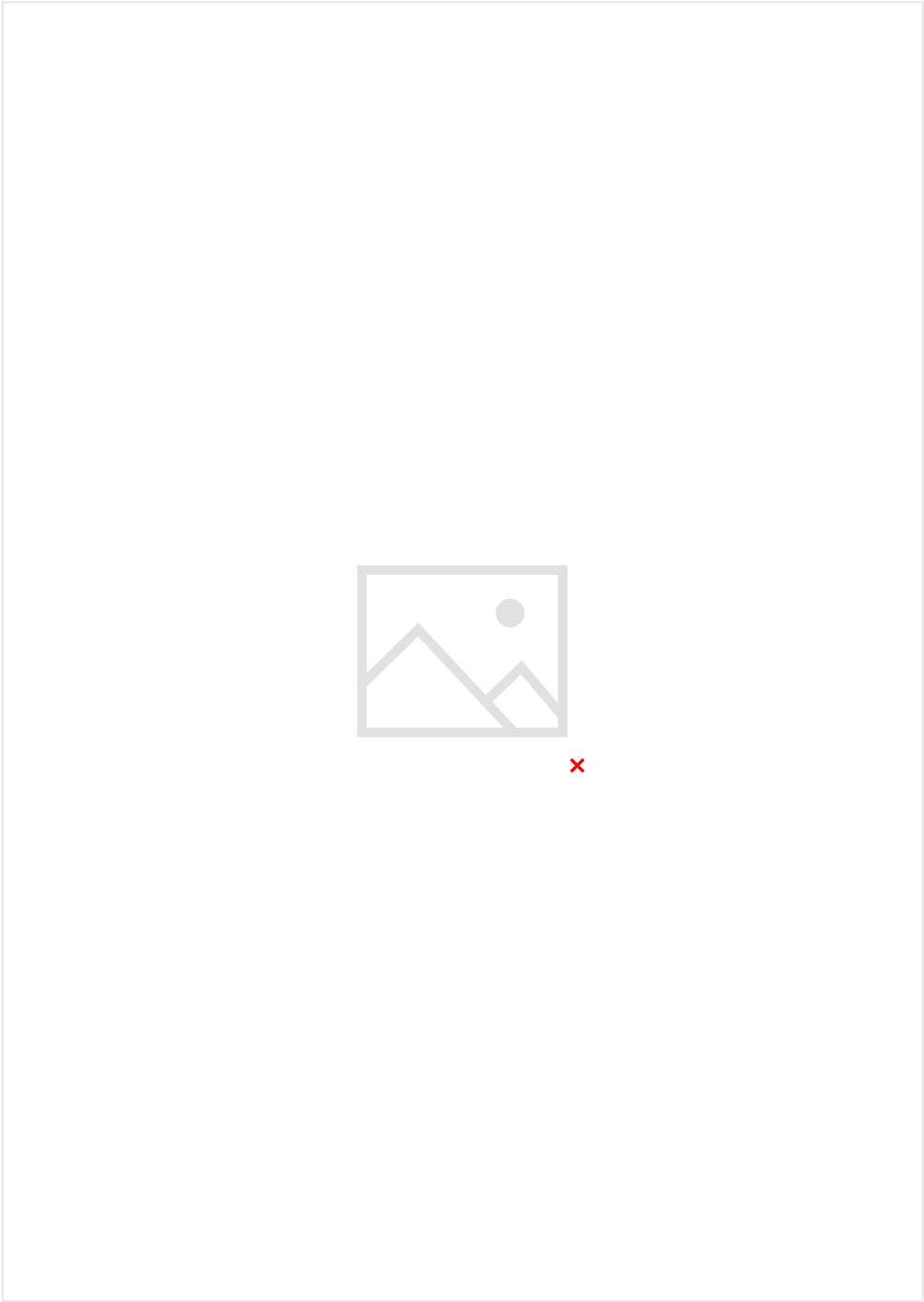
$$\frac{A_1}{(ps + q)} + \frac{A_2}{(ps + q)^2} + \cdots + \frac{A_m}{(ps + q)^m}$$

4. Quadratic Factors: For factor $(as^2 + bs + c)^n$, include the sum below:

$$\frac{B_1s + C_1}{(as^2 + bs + c)} + \frac{B_2s + C_2}{(as^2 + bs + c)^2} + \cdots + \frac{B_ns + C_n}{(as^2 + bs + c)^n}$$

5. Determine Unknowns: Equate original fraction to the sums found in steps 3 and 4, and solve for unknowns.







Quadratic Factors e.g.

Find: $\mathcal{L}^{-1}\left\{\frac{3}{s(s^2+2s+5)}\right\}$

For a quadratic of the form $as^2 + bs + c$, we can use the quadratic equation to determine its factors.

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For us, this gives complex factors, thus the term is irreducible.

$$s = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)} = -1 \pm 2j$$

$$\frac{3}{s(s^2 + 2s + 5)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 5}$$

Quadratic Factors e.g. - II

Basic Equation:

$$3 = A(s^2 + 2s + 5) + s(Bs + C) = (A + B)s^2 + (2A + C)s + 5A$$

For $s = 0$, we get: $3 = 5A$ thus $A = \frac{3}{5}$

Substituting A in gives: $3 = (\frac{3}{5} + B)s^2 + (\frac{6}{5} + C)s + 3$

Equating coefficients of powers of s gives:

$$(3 - 5A) = 0 \quad \text{and} \quad (6 - 5C) = 0$$

$$\left(\frac{3}{5} + B\right) = 0 \quad \text{and} \quad \left(\frac{6}{5} + C\right) = 0$$

We thus have $B = -\frac{3}{5}$ and $C = -\frac{6}{5}$.

We thus have:

$$\frac{3}{s(s^2 + 2s + 5)} = \frac{\frac{3}{5}}{s} + \frac{-\frac{3}{5}s - \frac{6}{5}}{s^2 + 2s + 5} = \frac{\frac{3}{5}}{s} + \left(-\frac{3}{5}\right) \frac{s + 2}{s^2 + 2s + 5}$$

Quadratic Factors e.g. - III

The quadratic term is the sum of the Laplace transforms of frequency shifted sine and cosine.

This gives:

$$\mathcal{L}\{K_1 e^{-at} \cos \omega t\} = \frac{K_1(s+a)}{(s+a)^2 + \omega^2} \quad \mathcal{L}\{K_2 e^{-at} \sin \omega t\} = \frac{K_2 \omega}{(s+a)^2 + \omega^2}.$$

Adding the transforms together, we get:

$$\mathcal{L}\{K_1 e^{-at} \cos \omega t + K_2 e^{-at} \sin \omega t\} = \frac{K_1(s+a) + K_2 \omega}{(s+a)^2 + \omega^2}$$

We next note that $(s+a)^2 + \omega^2 = s^2 + 2as + (a^2 + \omega^2)$

Equating to our denominator $(s^2 + 2s + 5)$, gives: $2a = 2$ and $(a^2 + \omega^2) = 5$

We thus have $a = 1$ and $\omega = 2$.

Quadratic Factors e.g. - IV

We now equate our numerators and get:

$$s + 2 = K_1(s + 1) + 2K_2 = K_1 s + (K_1 + 2K_2)$$

This gives us $K_1 = 1$ and $K_2 = \frac{1}{2}$

We thus have:

$$\frac{s+2}{s^2+2s+5} = \frac{(s+1) + (\frac{1}{2})2}{(s+1)^2 + 2^2}$$

Putting it all together:

$$\mathcal{L}^{-1}\left\{\frac{3}{s} + \left(-\frac{3}{5}\right)\frac{s+2}{s^2+2s+5}\right\} = \left[\frac{3}{5} - \frac{3}{5}e^{-t}(\cos 2t + \frac{1}{2}\sin 2t)\right]u(t)$$

We note that the factors of $s^2 + 2s + 5$ were $s = -1 \pm 2j$
 $= -a \pm \omega j$.

Stability Analysis Using Roots of $D(s)$

- The roots of $D(s)$ correspond to the exponential terms in the time domain response for $G(s)$, the transfer function.

$$\begin{aligned} G(s) &= \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \\ &= \frac{A_1}{(s+p_1)} + \frac{A_2}{(s+p_2)} + \cdots + \frac{A_n}{(s+p_n)} \end{aligned}$$

We refer to the roots of $D(s)$ as **poles**.

Solving for $g(t)$ gives:

$$\begin{aligned} g(t) &= \mathcal{L}^{-1}\left\{\frac{A_1}{(s+p_1)}\right\} + \mathcal{L}^{-1}\left\{\frac{A_2}{(s+p_2)}\right\} + \cdots + \mathcal{L}^{-1}\left\{\frac{A_n}{(s+p_n)}\right\} \\ &= A_1e^{-p_1t} + A_2e^{-p_2t} + \cdots + A_ne^{-p_nt} \end{aligned}$$

Stability Analysis: Real Roots

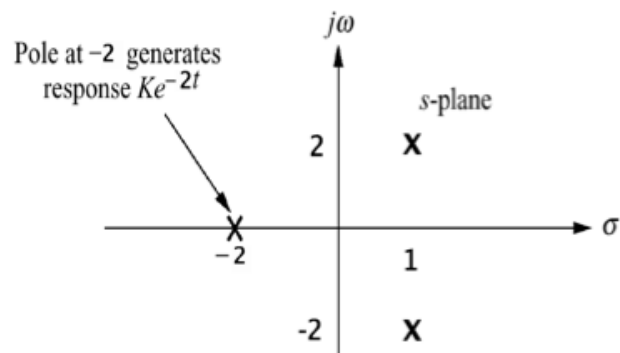
Considering: $g(t) = A_1 e^{-p_1 t} + A_2 e^{-p_2 t} + \dots + A_n e^{-p_n t}$

If we have pole(s) at $s = -p_i = -\sigma_i \pm j\omega_i$:

- If $\omega_i = 0$, then pole is strictly real and corresponds to $A_i e^{-\sigma_i t}$

If $\sigma_i > 0$, then pole in left side of imaginary plane, and response decreases to zero over time and system is **stable**.

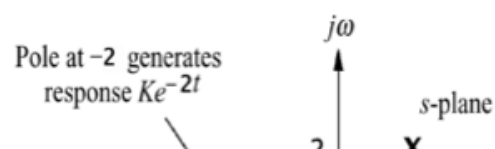
i.e. $\sigma_i = 2$, then we get
 $s = -2$, $p_i = 2$ and
 response
 $A_i e^{-\sigma_i t} = A_i e^{-2t}$.



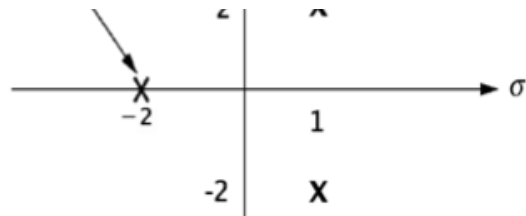
Stability Analysis: Real and Imaginary Roots

- If $\omega_i = 0$ and $\sigma_i < 0$, then pole in right side of imaginary plane, and response increases over time and system is **unstable**. i.e. $\sigma_i = -1$, then we get $s = 1$, $p_i = -1$ and response $A_i e^{-\sigma_i t} = A_i e^t$.
- If $\sigma_i = 0$ and $\omega_i \neq 0$, then we have two pure imaginary roots. This corresponds to a sinusoidal response with no damping, technically considered **stable**.

i.e. $\omega_i = 2$, then we get
 $s = \pm 2j$, $p_i = \pm 2j$ and



$s = \pm 2j$, $p_i = \pm 2j$ and
response
 $A_i \sin \omega_i t = A_i \sin 2t$.



Stability Analysis: Complex Roots

If we have $\sigma_i \neq 0$ and $\omega_i \neq 0$, we have complex roots.

- For poles at $s = -\sigma_i \pm j\omega_i$ we get the partial fraction expansion below:

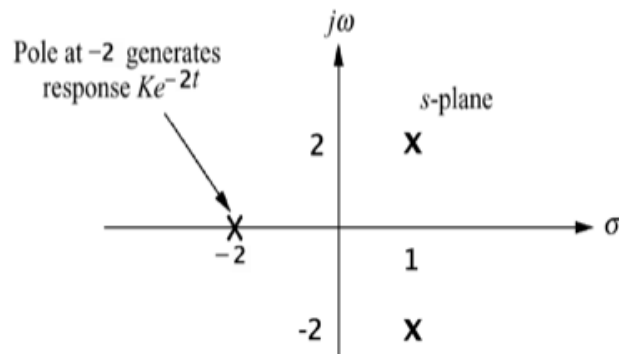
$$\frac{\alpha + j\beta}{s + \sigma_i + j\omega_i} + \frac{\alpha - j\beta}{s + \sigma_i - j\omega_i}$$

- This results in the time domain response of the form:

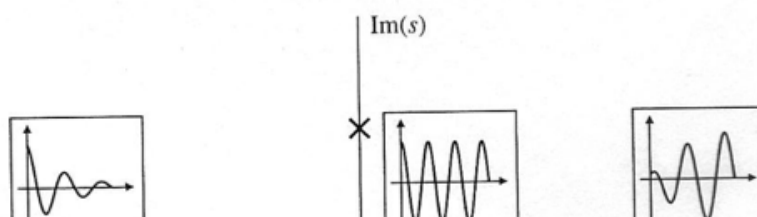
$$e^{-\sigma_i t} [2\alpha \cos \omega_i t + 2\beta \sin \omega_i t]$$

If $\sigma_i > 0$, then response is **stable**.

i.e. $\sigma_i = -2$ and $\omega_i = 3$,
then we get $s = 2 \pm 3j$,
 $p_i = -2 \pm 3j$ and response
 $e^{-\sigma_i t} [2\alpha \cos \omega_i t + 2\beta \sin \omega_i t]$
 $= e^{2t} [2\alpha \cos 3t + 2\beta \sin 3t]$.



Time Functions Associated with s -plane



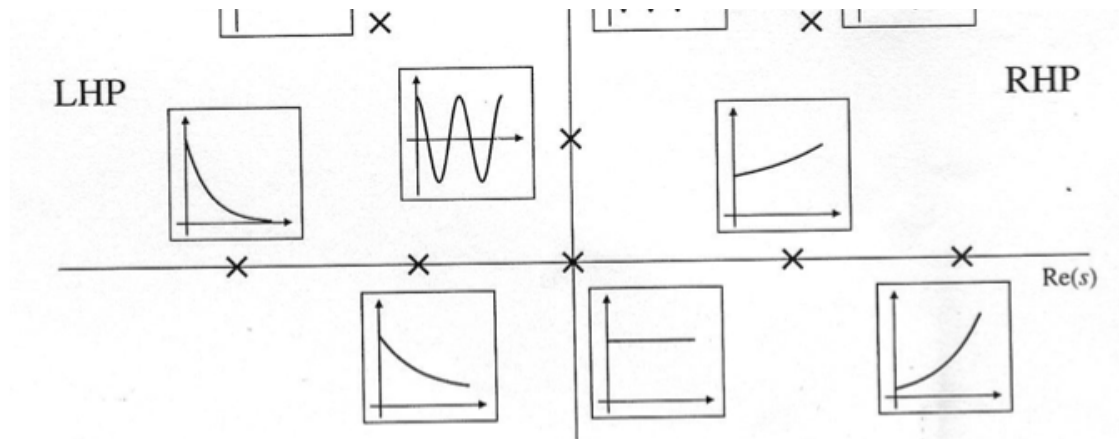
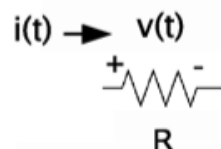


Figure 2.5 from Dorf and Bishop, *Modern Control Systems (10th Edition)*, Prentice-Hall, 2004.

Electric Network Transfer Function

- ▶ We will generalize notion of resistance to **impedance**.
- ▶ This will allow us to treat capacitors and inductors in similar manner as resistors in analyzing circuits.
- ▶ Recall that Ohms law says: $v(t) = Ri(t)$
- ▶ Taking Laplace Transform of both sides gives: $V(s) = RI(s)$
- ▶ We define the impedance to be $Z(s) = \frac{V(s)}{I(s)} = R$
- ▶ We also define the **admittance** to be: $Y(s) = \frac{I(s)}{V(s)} = \frac{1}{R} = G$

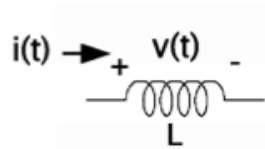


Impedance of Inductor

- ▶ The voltage-current relation for an inductor is: $v(t) = L \frac{di(t)}{dt}$.
- ▶ To find transfer function, take Laplace transform of each side and assume zero initial conditions:

$$V(s) = LsI(s)$$

- ▶ The impedance is thus $Z(s) = \frac{V(s)}{I(s)} = Ls$



Impedance of Capacitor

- ▶ The voltage-current relation for a capacitor is:

$$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$$

- ▶ Taking derivative of both sides gives:

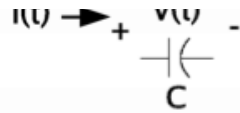
$$\frac{dv(t)}{dt} = \frac{1}{C} i(t)$$

- ▶ Taking Laplace transform (assuming zero initial conditions) gives:

$$sV(s) = \frac{1}{C} I(s)$$

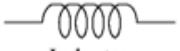
- ▶ The impedance is thus $Z(s) = \frac{V(s)}{I(s)} = \frac{1}{Cs}$

$$i(t) \rightarrow v(t)$$



Summary of Circuit Elements.

- ▶ Table 2.3 below shows for each element type the voltage-current, current-voltage, and voltage-charge relationship, as well as the elements impedance and admittance.
- ▶ All elements here are **passive** as they do not contain an internal energy source.

	$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$	$i(t) = C \frac{dv(t)}{dt}$	$v(t) = \frac{1}{C} q(t)$	$\frac{1}{Cs}$	Cs
Capacitor					
	$v(t) = Ri(t)$	$i(t) = \frac{1}{R} v(t)$	$v(t) = R \frac{dq(t)}{dt}$	R	$\frac{1}{R} = G$
Resistor					
	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_0^t v(\tau) d\tau$	$v(t) = L \frac{d^2 q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$
Inductor					

Note: The following set of symbols and units is used throughout this book: $v(t) = \text{V}$ (volts), $i(t) = \text{A}$ (amps), $q(t) = \text{Q}$ (coulombs), $C = \text{F}$ (farads), $R = \Omega$ (ohms), $G = \text{S}$ (mhos), $L = \text{H}$ (henries).

Equivalent Resistance and Impedance.

- ▶ It can be shown that resistance in serial can be replaced by an equivalent resistor with resistance R_s calculated as follows:

$$R_s = R_1 + R_2 + \cdots + R_N$$

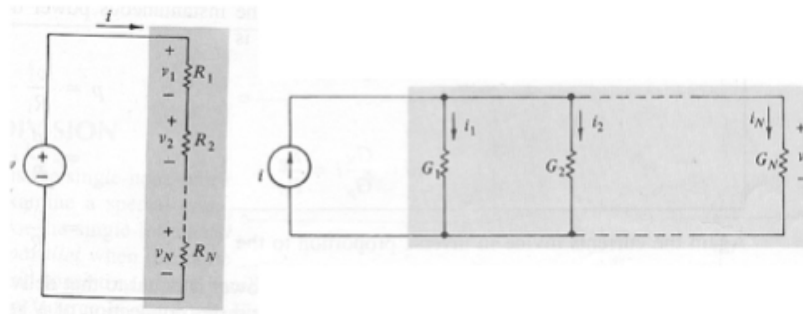
- ▶ This can be generalized to impedance as follows:

$$Z_s = Z_1 + Z_2 + \cdots + Z_N$$

- ▶ The equivalent resistance (R_p) and impedance (Z_p) when

elements are connected in parallel is:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N} \qquad \frac{1}{Z_p} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_N}$$



Figures 2.16 and 2.18 from *Electric Circuit Analysis*, D.E. Jounson et al., 1989.

Equivalent Resistance Example

Figure (a) to (b): $R_{s1} = 1 + 5 = 6\Omega$

Figure (b) to (c): $\frac{1}{R_p} = \frac{1}{12} + \frac{1}{6} \Rightarrow R_p = 4\Omega$

Figure (c) to (d): $R_{s2} = 7 + 4 = 11\Omega$

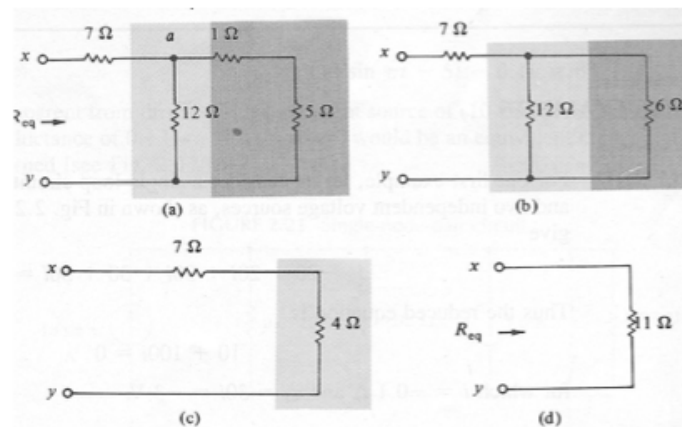


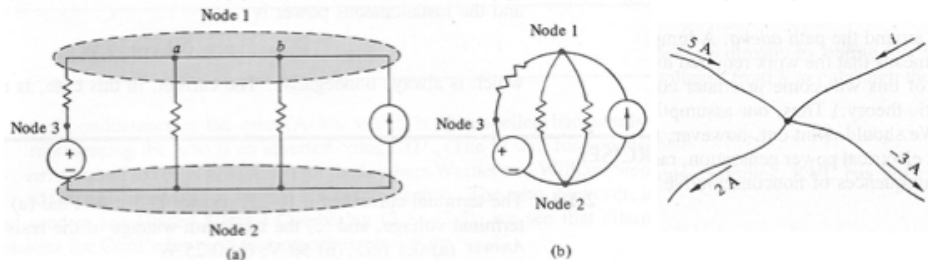
Figure 2.19 from *Electric Circuit Analysis*, D.E. Jounson et al., 1989.

Kirchhoff's Current Law

- A point of connection between two or more circuit elements is referred to as a **node**

referred to as a **node**.

- ▶ Circuit show in Fig. (a) contains three nodes and is electrically equivalent to that of Figure (b).
- ▶ **Kirchoff's current law (KCL)** says that *the algebraic sum of the currents entering any node is zero*.
- ▶ Currents flowing into the node are considered positive, and those leaving negative.
- ▶ For node on right, this would give: $5 + i - (-3) - 2 = 0$



Figures 2.8 and 2.9 from *Electric Circuit Analysis*, D.E. Jounson et al., 1989.

Kirchhoff's Voltage Law

- ▶ **Kirchoff's voltage law (KVL)** says that *the algebraic sum of voltages around any closed path is zero*.
- ▶ In the direction we traverse the path, voltages that go from $-$ to $+$ (lower to higher potential) are considered to be positive, and those going from $+$ to $-$ to be negative.
- ▶ Traversing the circuit below in clockwise direction gives:

$$5 - v - 10 + 2 = 0$$

- ▶ Solving for v gives $v = -3V$.

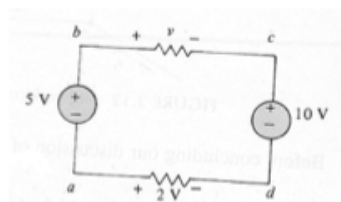


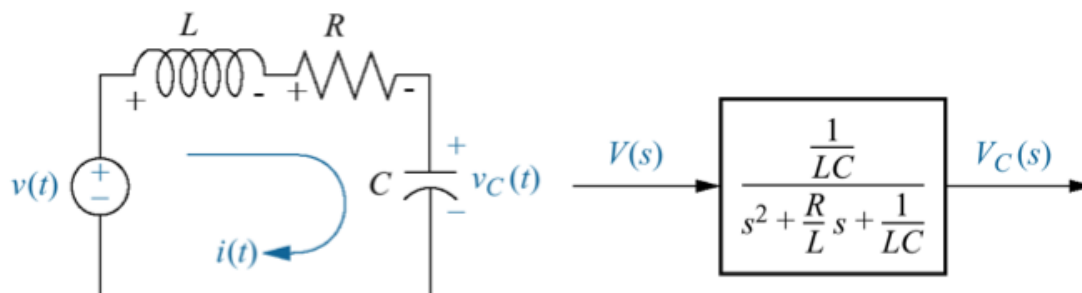
Figure 2.11 from *Electric Circuit Analysis*, D.E. Jounson et al., 1989.

关于KVL的方向问题：

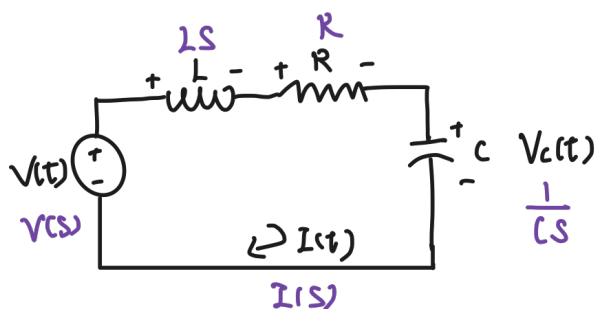
我们一般假设从电流从-到+流动，穿过一个电路元件的voltage drop是正的，从+流到-是负的。

Kirchhoff's Voltage Law eg.

- Find transfer function relation $\frac{V_C(s)}{V(s)}$.



Figures 2.4 and 2.5.



$$V(s) - V_L - V_R - V_C = 0$$

$$V(s) - LsI(s) - RI(s) - \frac{1}{Cs}I(s) = 0$$

$$V(s) - LsI(s) - RI(s) - \frac{1}{Cs}I(s) = 0$$

$$\text{Solving for } \frac{V_c(s)}{V(s)} = G(s)$$

$$V(s) = (Ls + R + \frac{1}{Cs})I(s)$$

$$I(s) = \frac{V(s)}{Ls + R + \frac{1}{Cs}}$$

$$V_c(s) = \frac{1}{Cs}I(s) = Z_c I(s)$$

$$V_c(s) = \frac{1}{Cs} \frac{V(s)}{Ls + R + \frac{1}{Cs}}$$

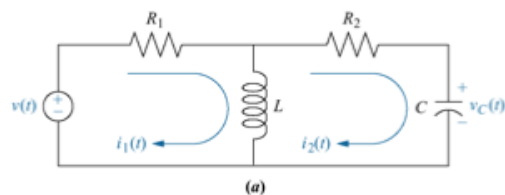
$$V_c(s) = \frac{V(s)}{Ls^2 + RCs + 1}$$

$$\frac{V_c(s)}{V(s)} = \frac{1}{Ls^2 + RCs + 1}$$

$$\frac{V_c(s)}{V(s)} = \frac{\frac{1}{2C}}{s^2 + \frac{R}{L}s + \frac{1}{2C}}$$

Mesh Analysis

- Replace passive elements with their impedances, and all sources and time variables with their Laplace transform.
- Assume a transform current and direction in each mesh.
- Assume for each element a voltage polarity
- Assume a current $I_3(s)$ and direction through the shared segment (ie. inductor)



- Apply Kirchoff's voltage law to each mesh going in direction of the mesh current.

- Use Kirchoff's current law to relate the currents.

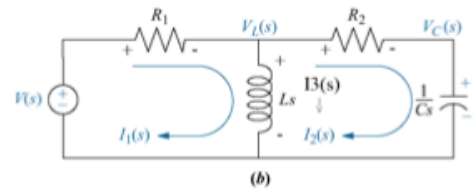


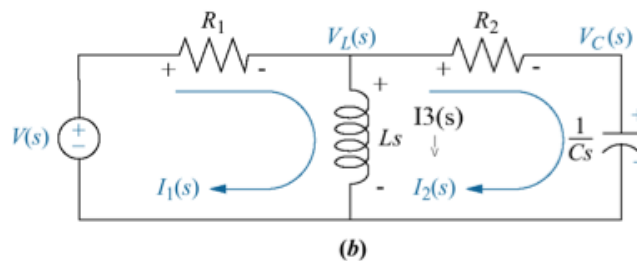
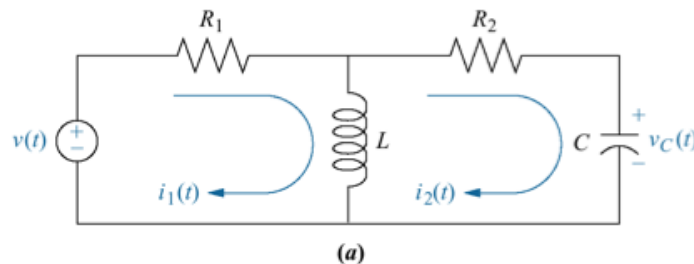
Figure 2.6.

Mesh analysis怎么用:

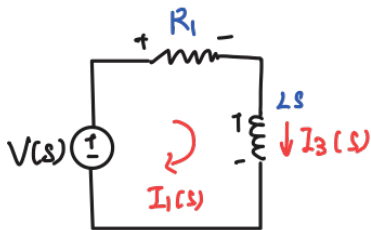
1. 先观察有几个闭合的loop，只取所有的最小闭合loop分配一个电流，方向都取顺时针。
2. 对于每个mesh，写出KVL表达式。
3. 对于mesh里面元器件的voltage drop，沿着电流方向走，写成 $-I \cdot Z$ ，如果两个mesh共享一个元器件，比如说以2.6举例子，那么L的voltage drop对于 I_1 而言就是 $-(I_1 - I_2) \cdot Ls$ ，对于 I_2 来说就是 $-(I_2 - I_1) \cdot Ls$ 。
4. 有几个mesh current就列几个方程，然后用Cramer's rule求解。
5. 找出想要的参数 (Laplace domain form) ，求出 $G(s)$ 。

Mesh Analysis eg.

- Find transfer function relation $\frac{I_2(s)}{V(s)}$.



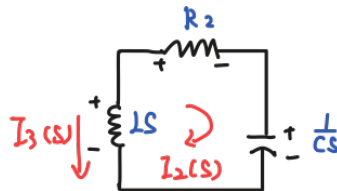
$$\frac{I_2(s)}{V(s)} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$



$$\text{KVL: } V(s) - R_1 I_1(s) - Ls I_3(s) = 0$$

$$V(s) - R_1 I_1(s) - Ls (I_1(s) - I_2(s)) = 0$$

$$\textcircled{1} (R_1 + Ls) I_1(s) - Ls I_2(s) = V(s)$$



KVL:

$$-R_2 I_2(s) - \frac{1}{Cs} I_2(s) - Ls (-I_3(s)) = 0$$

$$-R_2 I_2(s) - \frac{1}{Cs} I_2(s) - Ls [I_2(s) - I_1(s)] = 0$$

$$\textcircled{2} 0 = -Ls I_1(s) + (Ls + R_2 + \frac{1}{Cs}) I_2(s)$$

$$\therefore \begin{bmatrix} R_1 + Ls & -Ls \\ -Ls & Ls + R_2 + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$

$$\therefore I_2(s) = \frac{\begin{vmatrix} R_1 + Ls & V(s) \\ -Ls & 0 \end{vmatrix}}{\begin{vmatrix} R_1 + Ls & -Ls \\ -Ls & Ls + R_2 + \frac{1}{Cs} \end{vmatrix}} = \frac{0 + Ls V(s)}{(R_1 + Ls)(Ls + R_2 + \frac{1}{Cs}) - L^2 s^2}$$

$$= \frac{Ls V(s)}{R_1 Ls + R_1 R_2 + R_2 Ls + \frac{R_1}{Cs} + \frac{Ls}{Cs} \rightarrow (R_1 + R_2) Ls + \frac{1}{Cs} (R_1 + Ls) + R_1 R_2}$$

$$= \frac{V(s) Ls \cdot Cs}{(R_1 + R_2) LCs^2 + (R_1 + Ls) + R_1 R_2 Cs}$$

$$\therefore \frac{I_2(s)}{V(s)} = \frac{Ls^2}{LCs^2(R_1 + R_2) + (R_1 R_2 C + L)s + R_1}$$

$$(b) \frac{V_c(s)}{V(s)} \quad V_c(s) = I_2 Z_C$$

$$\underline{V_c(s)} = \underline{I_2(s)} \underline{V_c(s)}$$

$$\overline{V(s)} = \overline{V(s)} \cdot \overline{I_2(s)}$$

Cramer's Rule

Cramer's Rule: If $Ax = B$ is a system of n linear equations in n unknowns such that $\det(A) \neq 0$, then the system has a unique solution. The solution is:

$$x_1 = \frac{\det(A_1)}{\det(A)}, x_2 = \frac{\det(A_2)}{\det(A)}, \dots, x_n = \frac{\det(A_n)}{\det(A)}$$

where A_j is the matrix obtained by replacing the entries of the j^{th} column of A by the entries in the matrix B .

Definition taken from *Elementary Linear Algebra with Applications* by H. Anton et al. , 1987.

Matrix Determinant

The formula of a determinant for a 2×2 and a 3×3 matrix:

$$\begin{aligned} \text{(i) } \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} &= a_{11}a_{22} - a_{12}a_{21} \\ \text{(ii) } \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ &\quad - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} \end{aligned}$$

Definition taken from *Elementary Linear Algebra with Applications* by H. Anton et al. , 1987.

Mnemonic device for remembering determinant formula.

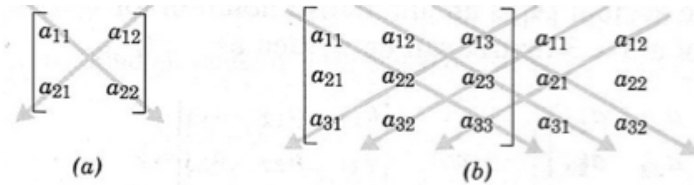
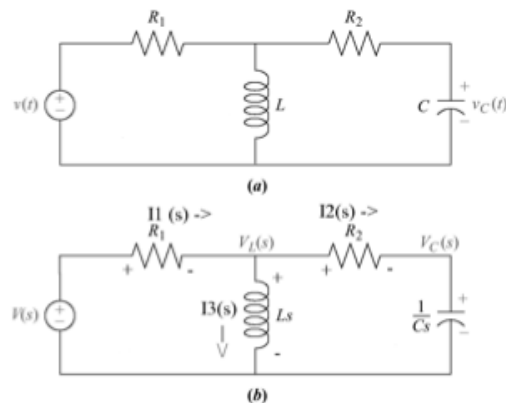


Figure 2.2 from *Elementary Linear Algebra with Applications* by H. Anton et al. , 1987.

Nodal Analysis

- ▶ Transform circuit into Laplace Domain.
- ▶ Assume for each element a voltage polarity.
- ▶ Determine nodes for circuit. Choose one (treat like ground) - others relative to it.
- ▶ Assume transform currents and directions entering and leaving each node.
- ▶ Use Kirchoff's current law to create equations for each node with an unknown voltage.
- ▶ Use relation $I(s) = \frac{V(s)}{Z(s)}$ to get equation for currents in terms of voltages.



Nodal analysis 怎么用:

1. 选定一个ground point作为0V参考点, 然后查看所有junction, 即有电流“汇合”和“分流”的地方, 给每个节点都分配一个参数 V_x 。
2. 对除了参考节点的每个节点, 进行KCL分析, 即sum流入=sum流出, 电流要写成 $\frac{V_x - V_y}{Z}$ 的形式。比如说, 对于2.6的例子, $I_1 = \frac{V(s) - V_L(s)}{R_1}$, $I_2(s) = \frac{V_L(s) - 0}{R_2 + 1/Cs}$, $I_3(s) = \frac{V_L(s) - 0}{Ls}$ 。注意: 如果该节点直接能判断出电压值, 比如说等于 V_s

或有V，就不需要给该节点加一个方程。

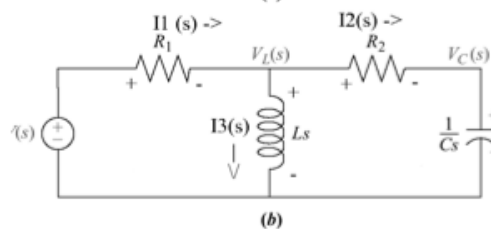
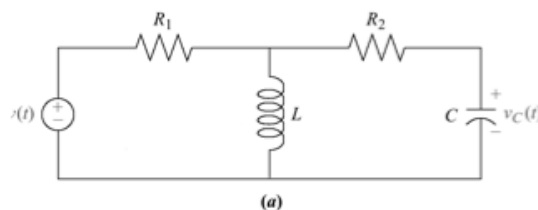
3. n个node有n-1个方程，求解方程得到每个node的含V(s)的表达式，然后通过欧姆定律算出来output port的V=I*Z，然后求出G(s)。

- 对于两种方法，都是用Laplace domain计算的。
- 写matlab的时候，syms里面要包含所有题目里面给的参数常量（比如V，L这些），以及我们设置的node（非参考点）和mesh current。

Matlab有一个特别好用的命令，叫equation to matrix；语法是 [A, b] = equationToMatrix([eqn1, eqn2], [var1, var2])

Nodal Analysis eg.

► Find transfer function relation $\frac{V_C(s)}{V(s)}$.



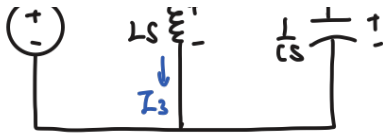
$$\frac{V_C(s)}{V(s)} = \frac{\frac{G_1 G_2}{C} s}{(G_1 + G_2)s^2 + \frac{G_1 G_2 L + C}{LC} s + \frac{G_2}{LC}}$$

Figures 2.6 and 2.7.

Ex. 5.



Nodal analysis



KCL:

$$I_1(s) - I_2(s) - I_3(s) = 0$$

$$I_1 = \frac{V - V_L(s)}{R_1}$$

$$I_2 = \frac{V_L(s)}{R_2 + \frac{1}{Cs}} \quad (1)$$

$$I_3 = \frac{V_L(s)}{Ls}$$

$$\therefore \frac{V_L(s) - V(s)}{R_1} + \frac{V_L(s)}{Ls} + \frac{V_L(s)}{R_2 + \frac{1}{Cs}} = 0 \quad (3)$$

sub (1) → (3), replace $I_2(s)$, solve for $V_L(s) = (4)$

sub (4) → (3), replace $V_L(s)$ solve for $\frac{V_L(s)}{V(s)}$

$$V_C(s) = \frac{1}{Cs} I_2(s) \quad (2)$$

Matlab verify:

```
syms S R1 R2 L C V VL
ZL = L*S;
ZC = 1/(C*S);
ZR2C = R2 + ZC;
I1 = (V - VL)/R1;
I2 = VL/ZR2C;
I3 = VL/ZL;
eq = I1 - I2 - I3 == 0;
VL_sol = simplify(solve(eq, VL));
VC = simplify(ZC * (VL_sol/ZR2C));
G=simplify(VC/V);
disp('Transfer Function G(s) = VC(s)/V(s):');
disp(G);
```

```
>> courseprac
Transfer Function G(s) = VC(s)/V(s):
(L*S)/(R1 + L*S + C*R1*R2*S + C*L*R1*S^2 + C*L*R2*S^2)
```

或者:

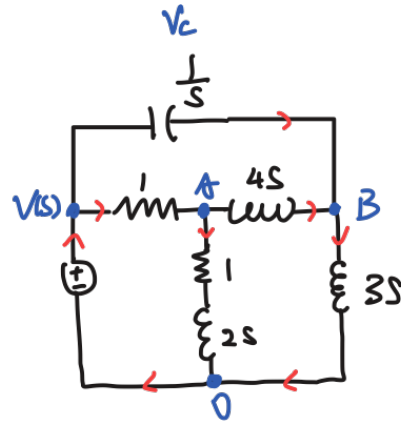
```
syms s V L R2 R1 C VL
eqn1=((VL-V)/R1)+(VL/(L*s))+(VL/(R2+(1/(C*s))))==0;
VL=solve(eqn1,VL);
VL=collect(VL);
I2=VL/(R2+(1/(C*s)));
VC=I2/(C*s);
G=collect(VC/V);
..
```

```

disp('The transfer function is:');
disp(G);
>> test
The transfer function is:
(L*s)/((C*L*R1 + C*L*R2)*s^2 + (L + C*R1*R2)*s + R1)

```

Ex. 6



Nodal analysis, Find $\frac{V_c(s)}{V(s)}$ 3 mesh loops
4 nodes

$V_c(s) = V(s) - V_B$; 2 unknowns $\rightarrow$ 2 equations

$$\textcircled{1} \quad \frac{V(s) - V_A}{1} = \frac{V_A - 0}{2s + 1} + \frac{V_A - V_B}{4s}$$

$$\textcircled{2} \quad \frac{V_A - V_B}{4s} + \frac{V(s) - V_B}{\frac{1}{s}} = \frac{V_B - 0}{3s}$$

$$\textcircled{1}: \quad V(s) - V_A = \frac{V_A}{2s + 1} + \frac{V_A - V_B}{4s}$$

$$\left(\frac{1}{2s + 1} + \frac{1}{4s} + 1\right) V_A - \frac{1}{4s} V_B = V(s)$$

$$\textcircled{2}: \quad \frac{V_B - V_A}{4s} + \frac{V_B}{3s} + s V_B = s V(s)$$

$$\left(s + \frac{1}{4s} + \frac{1}{3s}\right) V_B - \frac{1}{4s} V_A = s V(s)$$

$$\therefore \begin{bmatrix} \frac{1}{2s+1} + \frac{1}{7s} + 1 & -\frac{1}{4s} \\ -\frac{1}{4s} & \frac{V}{4s} + s + \frac{1}{3s} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} V(s) \\ sV(s) \end{bmatrix}$$

$$V_B = \frac{\begin{vmatrix} \frac{1}{2s+1} + \frac{1}{7s} + 1 & V(s) \\ -\frac{1}{4s} & sV(s) \end{vmatrix}}{\begin{vmatrix} \frac{1}{2s+1} + \frac{1}{7s} + 1 & -\frac{1}{4s} \\ -\frac{1}{4s} & \frac{V}{4s} + s + \frac{1}{3s} \end{vmatrix}}$$

$$\therefore V_C = V - V_B, \text{ find } \frac{V_C}{V} = \frac{V - V_B}{V}$$

```
syms s V
A = [ 1 + 1/(1+2*s) + 1/(4*s), -1/(4*s);
      -1/(4*s), s + 1/(4*s) + 1/(3*s) ];
b = [ V;
      s*V ];
A2 = A;
A2(:,2) = b;
VB = simplify(det(A2)/det(A));
Vc=V-VB;
G = simplify(Vc/V);
disp('G(s) =');
disp(G);
```

```
G(s) =
(8*s^2 + 13*s + 1)/(24*s^4 + 30*s^3 + 17*s^2 + 16*s + 1)
```

Operational Amplifiers

- ▶ So far we have only used passive elements for transfer functions.
- ▶ We can use **operational amplifiers (op amps)** to construct active circuits that can be used as transfer functions.
- ▶ Op amps have the following characteristics:

- Op amps have the following characteristics.

1. Output: $v_o(t) = A(v_2(t) - v_1(t))$
2. High input impedance: $Z_i = \infty$ (ideal)
3. Low output impedance: $Z_o = 0$ (ideal)
4. High constant gain: $A = \infty$ (ideal)

In linear operating region “ideal Op Amp” assumptions:

- (i) Input voltages are equal: $v_1(t) = v_2(t)$
- (ii) No current flows into inputs: $i_+ = i_- = 0$

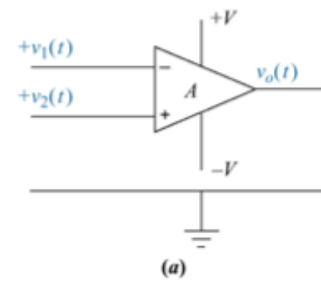


Figure 2.10.

Inverting Op Amps

- ▶ If we tie $v_2(t)$ to ground, we get an **inverting op amp**, with output $v_o(t) = -Av_1(t)$.
- ▶ Usually use with feedback in form of figure on the right.
- ▶ As input impedance very large, $I_a(s) \approx 0$.
- ▶ Using nodal analysis at v_1 , we find $I_1(s) = -I_2(s)$.
- ▶ If gain A very large, feedback forces $v_1(t) \approx 0$.

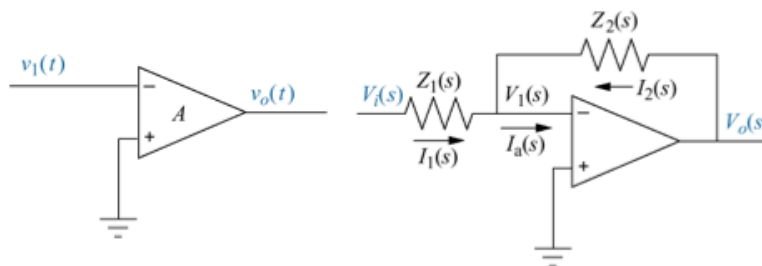


Figure2.10.

Inverting Op Amps - II

► With $v_1(t) \approx 0$, we thus have $I_1(s) = \frac{V_i(s)}{Z_1(s)}$ and $I_2(s) = \frac{V_o(s)}{Z_2(s)}$.

► Substituting into eqn $I_1(s) = -I_2(s)$ gives:

$$\frac{V_i(s)}{Z_1(s)} = -\frac{V_o(s)}{Z_2(s)}$$

► This gives the transfer function:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

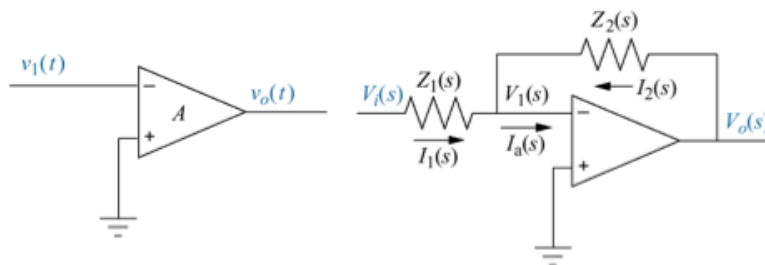


Figure2.10.

Inverting Op Amp eg.

► Find transfer function $\frac{V_o(s)}{V_i(s)}$.

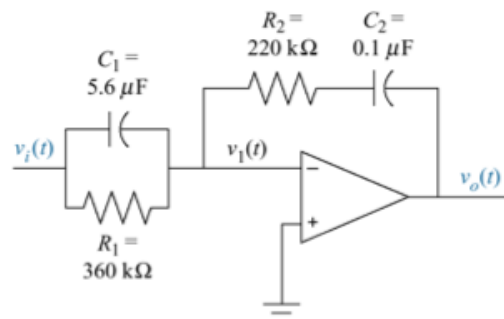


Figure 2.11.

Formula for inverting op amp is:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)} \quad (2)$$

We first need to determine $Z_1(s)$ and $Z_2(s)$ to use formula.

Using parallel inductance rule, we get:

$$\frac{1}{Z_1(s)} = \frac{1}{1/C_1s} + \frac{1}{R_1} = C_1s + \frac{1}{R_1}$$

Inverting Op Amp eg. - II

Which gives eqn below:

$$Z_1(s) = \frac{1}{C_1s + \frac{1}{R_1}} = \frac{1}{5.6 \times 10^{-6}s + \frac{1}{360 \times 10^3}} = \frac{360 \times 10^3}{2.016s + 1}$$

Using serial inductance rule, we get:

$$Z_2(s) = R_2 + \frac{1}{C_2s} = 220 \times 10^3 + \frac{10^7}{s}$$

Substituting into equation 2, we get:

$$\frac{V_o(s)}{V_i(s)} = - \frac{220 \times 10^3 + \frac{10^7}{s}}{\frac{360 \times 10^3}{2.016s + 1}}$$

Multiplying top and bottom by $(2.016s + 1)$ and simplifying gives:

$$- \frac{443,520s + 20,380,000 + \frac{10^7}{s}}{360 \times 10^3}$$

Inverting Op Amp eg. - III

$$- \frac{443,520s + 20,380,000 + \frac{10^7}{s}}{360 \times 10^3}$$

Multiply top and bottom by s and factoring out $\frac{443,520}{360 \times 10^3} = 1.232$ gives:

$$-1.232 \left[\frac{s^2 + 45.95s + 22.55}{s} \right]$$

We note that our equation is in the form of a PID controller as

we note that our equation is in the form of a PID controller, as shown below.

Transfer function of a PID controller:

$$G_c(s) = \frac{K_3(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3})}{s}$$

Noninverting Op Amp

► We know that $V_o(s) = A(V_i(s) - V_1(s))$

► Using voltage division, we can derive:

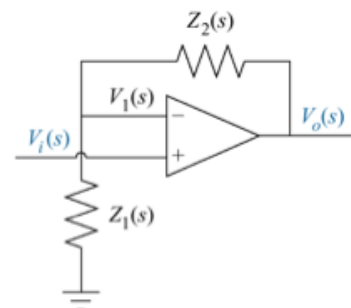
$$V_1(s) = \frac{Z_1(s)}{Z_1(s) + Z_2(s)} V_o(s)$$

► Substituting into top equation for $V_1(s)$, gives:

$$\frac{V_o(s)}{V_i(s)} = \frac{A}{1 + \frac{AZ_1(s)}{Z_1(s) + Z_2(s)}}$$

► As A is large, we can disregard the 1 term in denominator, which gives:

$$\frac{V_o(s)}{V_i(s)} = \frac{Z_1(s) + Z_2(s)}{Z_1(s)}$$



$$\begin{array}{l} \downarrow I_{z2} \\ \rightarrow I_- \\ \downarrow I_{z1} \end{array} \quad \begin{array}{l} I_{z2} - I_{z1} - I_- = 0 \\ \therefore I_{z2} = I_{z1} \end{array}$$

For ideal op-amp assumption: $I_+ = I_- = 0$

$$\therefore I_{z_1} = \frac{V_1(s) - 0}{z_1}, \quad I_{z_2} = \frac{V_0(s) - V_1(s)}{z_2}$$

$$\therefore \frac{V_0(s) - V_1(s)}{z_2} = -\frac{V_1(s)}{z_1}$$

Note: $V_1(s) = V_i(s)$

$$\therefore \frac{V_0(s) - V_i(s)}{z_2} = -\frac{V_i(s)}{z_1}$$

$$\frac{V_0(s)}{V_i(s)} = \frac{z_1 + z_2}{z_1}$$

Mechanical Systems

- ▶ We now examine how to relate the Laplace transform of the input to that of the output using a transfer function.
- ▶ We will consider two types of mechanical systems:
 1. **translational** mechanical systems.
 2. **rotational** mechanical systems.
- ▶ As the end result is a transfer function that is mathematically indistinguishable from that of an electrical network, we can thus connect the two by cascading their transfer functions.

Mass Component

- ▶ Newton's Second Law of Motion: $\Sigma f = Ma$
- ▶ We are interested in the following properties of the object:
 1. $\Sigma f = f(t)$: the summation of the forces acting on the object.
 2. M : the object's mass.
 3. $x(t)$: the position of the object.
 4. $v(t)$: the velocity of the object which is $\frac{dx}{dt}$.
 5. $a(t)$: the acceleration of the object which is $\frac{d^2x}{dt^2}$.
- ▶ We wish to find the transfer function $Z_m(s) = \frac{F(s)}{X(s)}$.
- ▶ We have: $f(t) = Ma(t) = M \frac{d^2x}{dt^2}$.
- ▶ Taking the Laplace transform of both sides gives: $F(s) = Ms^2X(s)$.
- ▶ Thus $Z_m(s) = \frac{F(s)}{X(s)} = Ms^2$.
- ▶ Similarly, $Z'_m(s) = \frac{F(s)}{V(s)} = Ms$

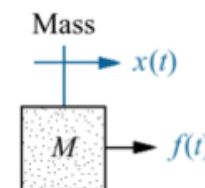


Table 2.4

Viscous Damper

- ▶ Middle part is typically moving through some sort of fluid.
- ▶ Frictional force generated is proportional to object's velocity:

$$f(t) = f_v v(t)$$

where f_v is called *coefficient of viscous friction*.

- ▶ Substituting $v(t) = \frac{dx}{dt}$ gives

$$f(t) = f_v \frac{dx}{dt}$$

- ▶ Taking the Laplace transform of both sides gives:
 $F(s) = f_v sX(s)$.

- ▶ Thus $Z_m(s) = \frac{F(s)}{X(s)} = f_v s$.
- ▶ Similarly, $Z'_m(s) = \frac{F(s)}{V(s)} = f_v$

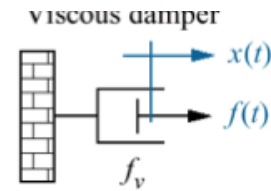


Table 2.4.

Spring Component

- ▶ The *spring constant* is denoted K .
- ▶ Spring's force is proportional to distance x which gives us:

$$f(t) = Kx(t)$$

- ▶ Taking the Laplace transform of both sides gives:

$$F(s) = KX(s)$$

- ▶ We thus have: $Z_m(s) = \frac{F(s)}{X(s)} = K$.
- ▶ With zero initial conditions, we have:
 $x(t) = \int_0^t v(\tau) d\tau$
- ▶ Thus $F(s) = \frac{K}{s} V(s)$
- ▶ This gives: $Z'_m(s) = \frac{F(s)}{V(s)} = \frac{K}{s}$

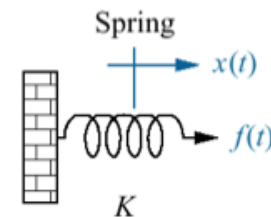
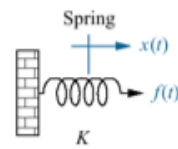
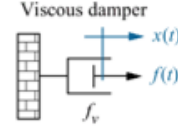
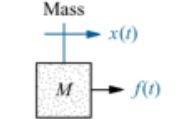


Table 2.4.

Summary of Translational Elements

- ▶ Table gives Force-velocity, force-displacement, and impedance translational relationships for springs, viscous dampers, and mass.

Component	Force-velocity	Force-displacement	Impedance $Z_m(s) = F(s)/X(s)$
-----------	----------------	--------------------	-----------------------------------

	$f(t) = K \int_0^t v(\tau) d\tau$	$f(t) = Kx(t)$	K
	$f(t) = f_v v(t)$	$f(t) = f_v \frac{dx(t)}{dt}$	$f_v s$
	$f(t) = M \frac{dv(t)}{dt}$	$f(t) = M \frac{d^2 x(t)}{dt^2}$	Ms^2

Note: The following set of symbols and units is used throughout this book: $f(t)$ = N (newtons), $x(t)$ = m (meters), $v(t)$ = m/s (meters/second), K = N/m (newtons/meter), f_v = N-s/m (newton-seconds/meter), M = kg (kilograms = newton-seconds²/meter).

Table 2.4.

Translational eg. - one equation of motion

- ▶ Find transfer function $\frac{X(s)}{F(s)}$
- ▶ $f(t)$ is the force we are applying in the direction of $x(t)$.
- ▶ To solve for transfer function, do the following:
 1. Draw free body drawing showing all forces acting on object, and their directions.
 2. Replace mechanical components with mechanical impedances.
 3. Newton's law says that $Ma(t)$ equals the sum of the forces acting on the object. Use this to create your force equation.

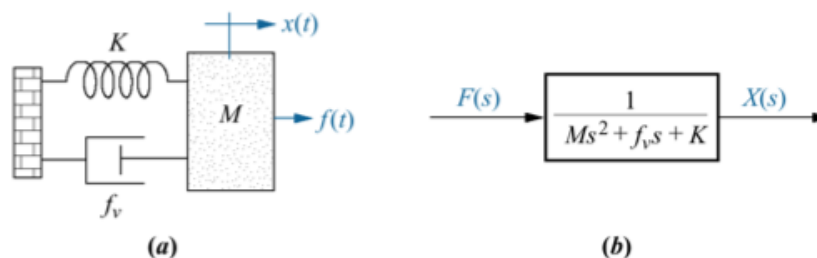


Figure 2.15.

建模思路：

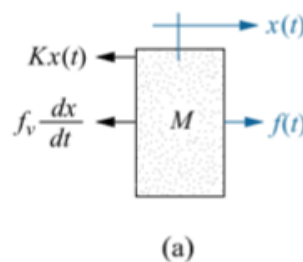
1 首先 确定正移动方向 这很重要· 找出degree of freedom (检查任何两个弹性

1. 首先，确定正移动方向，这很重要，找出degree of freedom (但三位内部1并位物件 (比如spring, damper) 之间的物体，即使没有质量，注意那种竖着接进来的不算)，所有计算都在Laplace domain进行。
2. 分析每个物体的时候，首先假设除了该物体在运动，其他物体均静止不动，然后分析该物体的受力，注意不要忘了假想的加速度力，方向和移动正方向相反。注意，此时我们的位移是该物体的位移。
3. 然后假设该物体静止，所有与之相连接的物体运动，分析这些运动的物体给我们被分析的物体施加的力；注意，此时受力方向是和连接到被分析物体，即此时正在运动的外部物体的运动方向，并且位移是外部物体的位移。
4. 把equation整理成 $Ax=b$ ，然后用Cramer's rule求解。

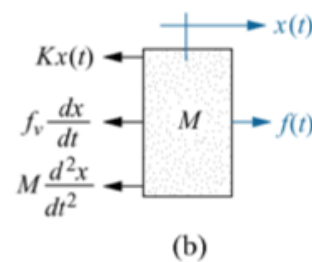
Translational eg. - one equation of motion - II

- ▶ We assume the mass is travelling to the right.
- ▶ This means only the applied force ($f(t)$) points to the right.
- ▶ The spring and viscous damping forces impede the motion, thus act to oppose it.
- ▶ Newton's second law: $M \frac{d^2x}{dt^2} = f(t) - f_v \frac{dx}{dt} - Kx(t)$
- ▶ Rearranging this gives: $f(t) = M \frac{d^2x}{dt^2} + f_v \frac{dx}{dt} + Kx(t)$

- ▶ (a) Proper application of Newton's law.



- ▶ (b) Text book's "method."

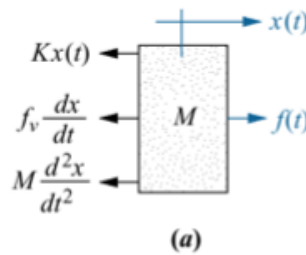


Free body diagram..

Translational eg. - one equation of motion - III

- ▶ Can think of the $Ma(t)$ term as force beyond spring and viscous damping forces, needed to attain acceleration $a(t)$.
- ▶ Using textbook's method, forces acting in assumed direction of motion are positive, those acting in opposite are negative, and they should sum to zero.

- ▶ (a) Free-body diagram of mass, spring, and damper system.



- ▶ (b) Transformed free-body diagram.

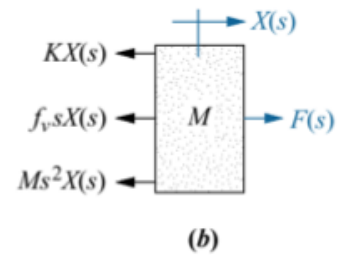
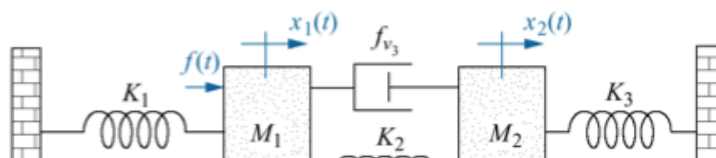


Figure 2.16.

其实ma不应该被视作一个“外力”，但是我们为了方便推导transfer function，我们画受力分析图的时候也顺手画上了。

Translational eg. - two degrees of freedom

- ▶ Number of equations of motion needed = number of **linearly independent motions**.
- ▶ A linearly independence means that one object can still move when all other objects are held still.
- ▶ We also refer to number of linearly independent motions as the number of **degrees of freedom** of the system.



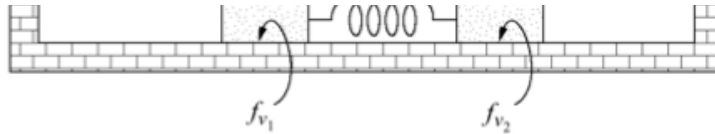


Figure 2.17 (a).

Translational eg. - two degrees of freedom - II

- ▶ Find transfer function $\frac{X_2(s)}{F(s)}$
- ▶ To solve a problem like this, we will use the principal of **superposition**. To find the forces acting on a given object (say M_1), we do the following:
 1. Draw the free body diagram for object (ie. M_1) by holding all other objects (ie. M_2) still, and finding the forces acting on the object (ie. M_1) due only to its motion.
 2. We then hold the object (ie. M_1) still, and activate each other object (ie. M_2) one at a time, and determine the forces acting on original object (ie. M_1) due to the other object's motion.
 3. The total force acting on the original object is the superposition of the forces found in steps 1 and 2.

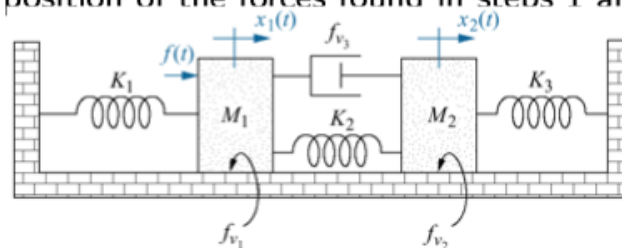


Figure 2.17 (a).

简单来说就是，假设我们现在分析M1，我们先假设所有其他的物件都不动，只有M1移动，此时M1受力情况是什么样，然后假设M1不动，只有M2运动，再分析M1的受力，最后叠加。

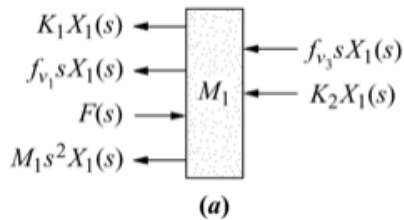
关于方向：当分析的物体自身在运动的时候，我们一般把friction, damper force都设置为和速度方向相反，spring force和位移方向相反，而我们的假想惯性力（加速度力）和位移方向相反。

对于被分析物体自身静止分析其他物体运动给他造成的影响的时候，被分析物体受到的外力的方向和正在移动的外部物体的移动方向相同。

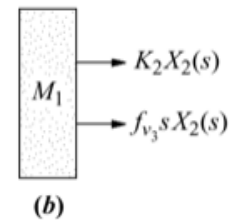
Translational eg. - two degrees of freedom - III

- Use Newton's law to find equation of motion for M_1 .

- (a) Forces on M_1 due to only motion of M_1 .



- (b) Forces on M_1 due to only motion of M_2 .



- (c) All forces on M_1 .

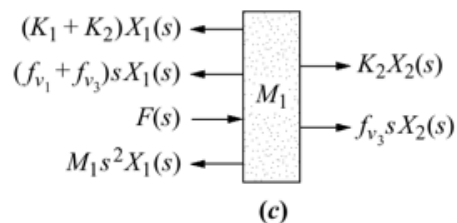
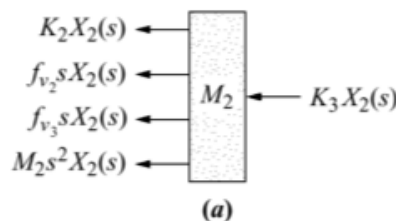


Figure 2.18.

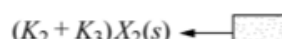
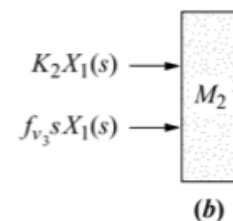
Translational eg. - two degrees of freedom - IV

- Use Newton's law to find equation of motion for M_2 .

- (a) Forces on M_2 due to only motion of M_2 .



- (b) Forces on M_2 due



to only motion of M_1 .

- (c) All forces on M_2 .

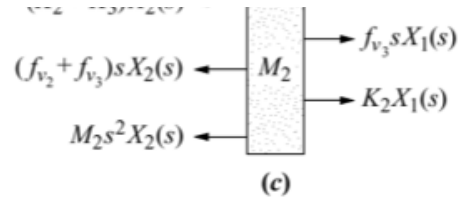


Figure 2.19.

$$(k_1 + k_2)x_1 + (f_{v1} + f_{v2})sX_1 + M_1s^2X_1 = F(s) + k_2X_2 + f_{v3}sX_2$$

$$[M_1s^2 + (f_{v1} + f_{v2})s + k_1 + k_2]X_1 - (f_{v3}s + k_2)X_2 = F(s)$$

$$(k_2 + k_3)X_2 + (f_{v2} + f_{v3})sX_2 + M_2s^2X_2 = f_{v3}sX_2 + k_2X_1$$

$$- (f_{v3}s + k_2)X_1 + [M_2s^2 + (f_{v2} + f_{v3})s + k_2 + k_3]X_2 = 0$$

$$\therefore Ax = b$$

$$A = \begin{bmatrix} M_1s^2 + (f_{v1} + f_{v2})s + k_1 + k_2 & - (f_{v3}s + k_2) \\ - (f_{v3}s + k_2) & M_2s^2 + (f_{v2} + f_{v3})s + k_2 + k_3 \end{bmatrix}$$

$$x = \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} \quad b = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$\frac{X_2(s)}{F(s)}, \quad X_2(s) = \frac{|A_2|}{|A|}$$

$$A_2 = \begin{bmatrix} M_1s^2 + (f_{v1} + f_{v2})s + k_1 + k_2 & F(s) \\ - (f_{v3}s + k_2) & 0 \end{bmatrix}$$

$$|A_2| = (f_{v3}s + k_2)F(s)$$

$$|A| = [m_1 s^2 + (f_{v1} + f_{v2})s + k_1 + k_2] [m_2 s^2 + (f_{v2} + f_{v3})s + k_2 + k_3] - (f_{v2} s + k_2)^2$$

$$\frac{x_2}{F} = \frac{f_{v2} s + k_2}{[m_1 s^2 + (f_{v1} + f_{v2})s + k_1 + k_2] [m_2 s^2 + (f_{v2} + f_{v3})s + k_2 + k_3] - (f_{v2} s + k_2)^2}$$

注意，在mechanical system里面impedance没有任何意义，只是当计算Laplace的时候，微分方程会变成 $T(s)=Z(s)*\theta(s)$ 的形式， $Z(s)=Js^2+Ds+K$ 的形式，方便整理。

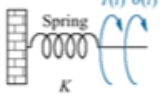
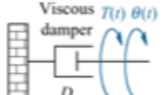
Rotational Mechanical Systems

- ▶ With rotational mechanical systems, we are dealing with objects that rotate about a fixed axis.
- ▶ The main differences from a translational system are as follows:
 - ▶ $x(t)$ displacement becomes $\theta(t)$ **angular displacement**.
 - ▶ $v(t) = \frac{dx(t)}{dt}$ velocity becomes $\omega(t) = \frac{d\theta(t)}{dt}$ **angular velocity**.
 - ▶ $f(t)$ force becomes $T(t)$ **torque**.
 - ▶ M mass becomes J **moment of inertia**.
 - ▶ f_v coefficient of viscous damping becomes D **coefficient of viscous angular damping**.
 - ▶ K spring constant becomes K **angular spring constant**.

Rotation system和translational类似，只是把质量换成moment of inertia J

Summary of Rotational Components

- ▶ Same symbols for components, but undergoing rotation.
- ▶ We are interested in the relation: $T(s) = Z_m(s)\theta(s)$

Component	Torque-angular velocity	Torque-angular displacement	Impedance $Z_u(s) = T(s)/\theta(s)$
 Spring K	$T(t) = K \int_0^t \omega(\tau) d\tau$	$T(t) = K\theta(t)$	K
 Viscous damper D	$T(t) = D\omega(t)$	$T(t) = D \frac{d\theta(t)}{dt}$	Ds
 Inertia J	$T(t) = J \frac{d\omega(t)}{dt}$	$T(t) = J \frac{d^2\theta(t)}{dt^2}$	Js^2

Note: The following set of symbols and units is used throughout this book: $T(t)$ = N-m (newton-meters), $\theta(t)$ = rad (radians), $\omega(t)$ = rad/s (radians/second), K = N-m/rad (newton-meters/radian), D = N-m-s/rad (newton-meters-seconds/radian), J = kg-m² (kilogram-meters² = newton-meters-seconds²/radian).

Table 2.5.

Would have for $Z'_m(s) = \frac{T(s)}{\omega(s)}$ the values $\frac{K}{s}$, D , and $J s$.

Degrees of Freedom

- ▶ Similar idea as for translation systems.
- ▶ The **degrees of freedom** is the number of objects (*points of motions*) that can still rotate when all other objects are held still.
- ▶ The degrees of freedom equals the number of equations of motion we need to describe a system.

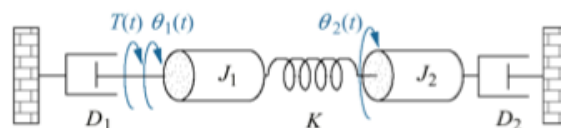


Figure 2.22 (b).

注意，Kinematics和Dynamics的DOF语境不一样，kinematics就是课上讲的，只要能够独立的转，就算一个自由度，但是dynamics要求必须有m或者J。

Writing Equations of Motion - Rotational

- ▶ Create free body diagram, but show torques and angular displacement instead.
- ▶ We will use superposition. To find the torques action on J_1 :
 1. Draw the free body diagram for J_1 by holding J_2 still, and find the torques acting on J_1 due only to its own motion.
 2. We then hold J_1 still, and rotate J_2 , and determine the torques acting on J_1 due to the motion of J_2 .
 3. The total torque acting on J_1 is the superposition of the torques found in steps 1 and 2.
- ▶ Repeat above, but for J_2 .
- ▶ The sum of torques acting in direction of assumed rotation equals the sum of the torques opposing.

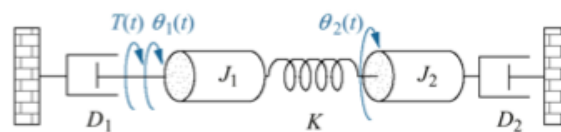


Figure 2.22 (b).

这里我们假设加速度的方向和转动方向相反，这只是一个人为参考系。这里的写法只是调整了顺序，在物理课上我们习惯把 $J^*a = \Sigma \tau$ 。

Rotational eg.

- ▶ Find the transfer function $\frac{\theta_2(s)}{T(s)}$.
- ▶ We have a flexible rod being supported at both ends by bearings, and is undergoing torsion.
- ▶ We assume system can be approximated by a spring at one point in the rod, with an inertia of J_1 on the left, and J_2 on the right.

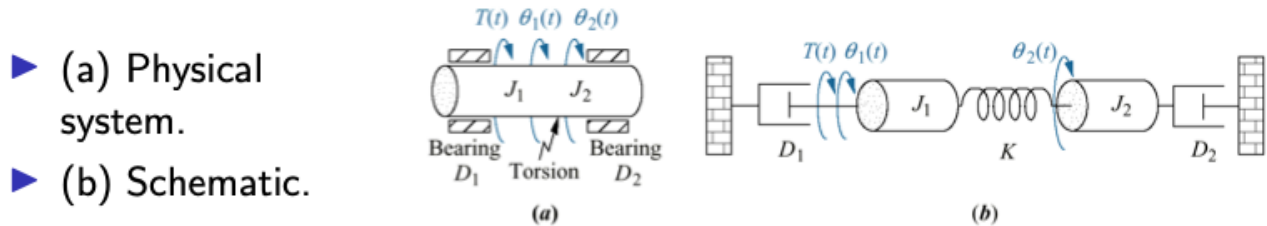


Figure 2.22.

Free body Diagrams for J_1

- ▶ (a) Torques on J_1 due only to motion of J_1 .
- ▶ (b) Torques on J_1 due only to motion of J_2 .
- ▶ (c) All Torques on J_1 .

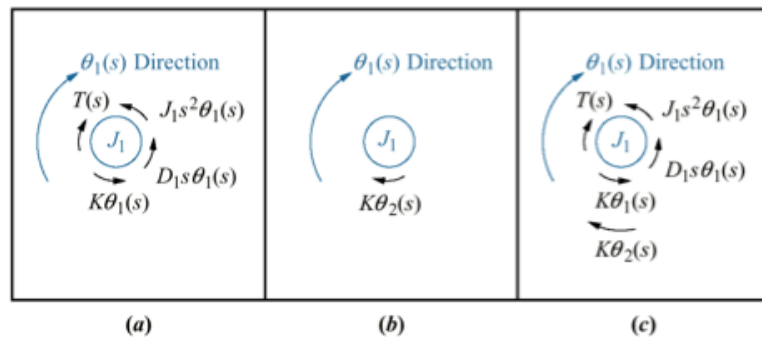


Figure 2.23.

We now sum torques opposing rotation and equate that to the torques in direction of rotation giving:

$$J_1 s^2 \theta_1(s) + D_1 s \theta_1(s) + K \theta_1(s) = T(s) + K \theta_2(s) \quad (3)$$

Free body Diagrams for J_2

- ▶ (a) Torques on J_2 due only to motion of J_2 .

- ▶ (b) Torques on J_2 due only to motion of J_1 .

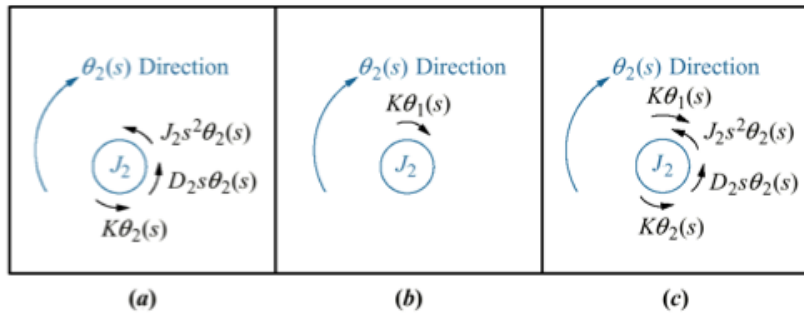


Figure 2.24.

- ▶ (c) All torques on J_2 .

We now sum torques opposing rotation and equate that to the torques in direction of rotation giving:

$$J_2 s^2 \theta_2(s) + D_2 s \theta_2(s) + K \theta_2(s) = K \theta_1(s) \quad (4)$$

Rotational eg. - II

For equations 3 and 4, we move unknowns to the left side, and collect terms for θ_1 and θ_2 giving:

$$(J_1 s^2 + D_1 s + K) \theta_1(s) - K \theta_2(s) = T(s) \quad (5)$$

$$-K \theta_1(s) + (J_2 s^2 + D_2 s + K) \theta_2(s) = 0 \quad (6)$$

Putting this in the form of $\underline{A} \underline{x} = \underline{B}$ gives:

$$\begin{bmatrix} J_1 s^2 + D_1 s + K & -K \\ -K & J_2 s^2 + D_2 s + K \end{bmatrix} \begin{bmatrix} \theta_1(s) \\ \theta_2(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

$$\underline{x} = \begin{bmatrix} \theta_1(s) \\ \theta_2(s) \end{bmatrix}; \quad \underline{A} = \begin{bmatrix} J_1 s^2 + D_1 s + K & -K \\ -K & J_2 s^2 + D_2 s + K \end{bmatrix}$$

$$\underline{B} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix};$$

Rotational eg. - III

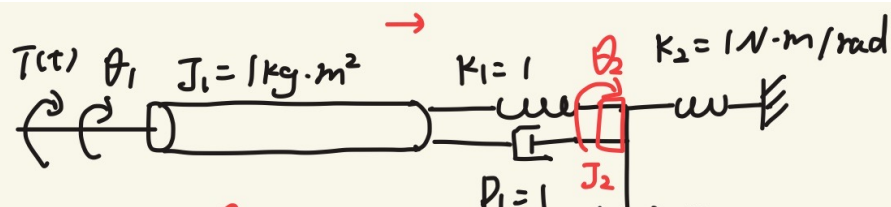
Cramer's rule given below where $\underline{A}_2$ is $\underline{A}$ with second column replace with $\underline{B}$:

$$\theta_2(s) = \frac{|\underline{A}_2|}{|\underline{A}|} = \frac{\left| \begin{bmatrix} J_1 s^2 + D_1 s + K & T(s) \\ -K & 0 \end{bmatrix} \right|}{|\underline{A}|} = \frac{KT(s)}{|\underline{A}|}$$

We thus have:

$$\frac{\theta_2(s)}{T(s)} = \frac{K}{|\underline{A}|}$$

独立旋转的标准：有没有被两个弹性元件或者damper隔开。



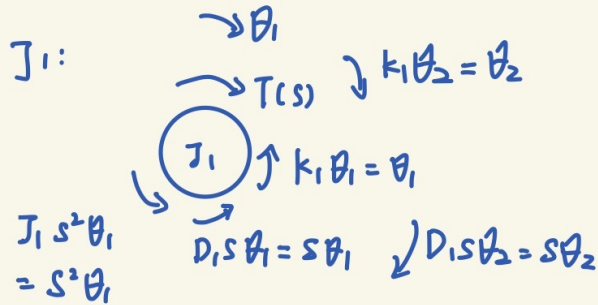
$$J_2 = 0$$

$$\boxed{1} \quad D_2 = 1 \text{ Nms/rad}$$

2 degrees of freedom ,

2 equations, 2 unknowns

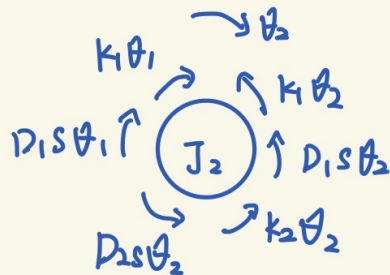
$$\text{Want } G(s) = \frac{\theta_2(s)}{T(s)}$$



$$\therefore s^2 \theta_1 + s \theta_1 + \theta_1 = T(s) + s \theta_2 + \theta_2$$

$$(s^2 + s + 1) \theta_1 - (s + 1) \theta_2 = T(s) \quad (1)$$

J_2 :



$$2s \theta_2 + 2 \theta_2 = s \theta_1 + \theta_1$$

$$-(s + 1) \theta_1 + 2(s + 1) \theta_2 = 0 \quad (2)$$

$$\theta_2(s) = \frac{|A_2|}{|A|} \quad , \quad x = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad , \quad b = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2(s + 1) \end{bmatrix}$$

$$A_2 = \begin{bmatrix} s^2 + s + 1 & T \\ -(s+1) & 0 \end{bmatrix}$$

$$\det(A_2) = (s+1)T$$

$$\det A = (s^2 + s + 1)[2(s+1)] - (s+1)^2$$

$$\theta_2 = \frac{(s+1)T}{(s+1)[2s^2 + 2s + 2 - s - 1]}$$

$$\frac{\theta_2}{T} = G(s) = \frac{1}{2s^2 + s + 1}$$

Rotational Mechanical System with Gears

- Usually when rotational systems are driven by motors, we find associated gear trains driving the load.
- In diagram below, we see that if we apply torque $T_1(t)$ to our

input gear which has N_1 teeth and radius r_1 , we will get a corresponding torque $T_2(t)$ out of gear 2, which has N_2 teeth, and radius r_2 .

- We want to determine the ratios $\frac{\theta_2}{\theta_1}$ and $\frac{T_2}{T_1}$.

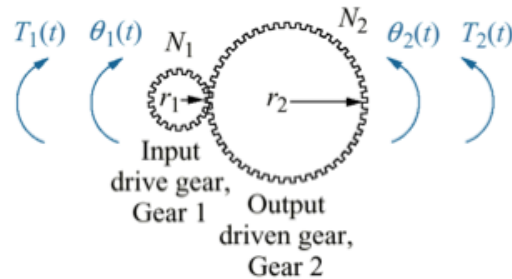


Figure 2.27.

Transfer Functions for Gears

- When gears turn, distance travelled around each gear's circumference is equal.
- We thus have: $r_1\theta_1 = r_2\theta_2 \Rightarrow \frac{\theta_2}{\theta_1} = \frac{r_1}{r_2}$
- It can be shown that $\frac{r_1}{r_2} = \frac{N_1}{N_2}$
- We thus have: $\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2}$

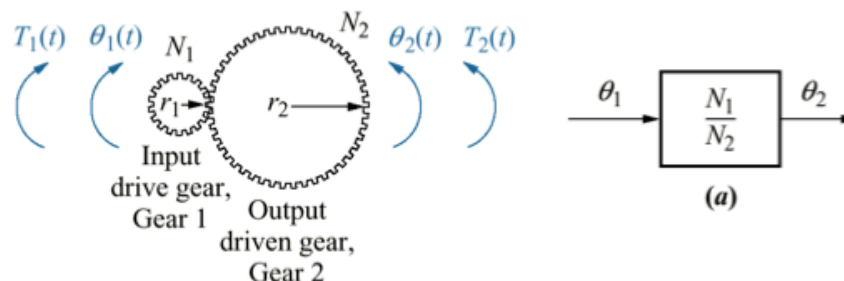


Figure 2.27 and Figure 2.28.

Transfer Functions for Gears - lossless

- ▶ If we assume the gears have negligible inertia and damping, then the energy put into gear 1 equals energy out of gear 2.
- ▶ We thus have: $T_1\theta_1 = T_2\theta_2 \Rightarrow \frac{T_2}{T_1} = \frac{\theta_1}{\theta_2}$
- ▶ Substituting in transfer function: $\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2}$
- ▶ We get: $\frac{T_2}{T_1} = \frac{N_2}{N_1}$

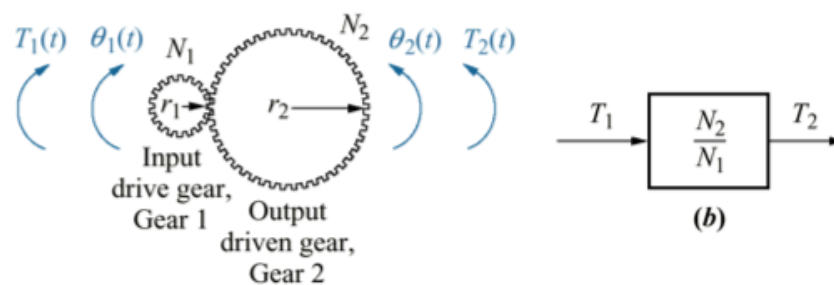


Figure 2.27 and Figure 2.28.

Using Transfer Function for Lossless Gears

- ▶ Our first step is to reflect T_1 to the other side of output gear giving: $T_1(s) \frac{N_2}{N_1}$.
- ▶ This gives equation of motion:

$$(Js^2 + Ds + K)\theta_2(s) = T_1(s) \frac{N_2}{N_1}$$

- ▶ This would allow us to find: $\frac{\theta_2(s)}{T_1(s)}$





Figure 2.29.

这里只有一个degree of freedom，因为gear是independent的。

注意：我们一般把齿轮关系只做数值等效，不会因为他们的转动方向相反就乘以-1，因为这样很麻烦，我们的做法是对于不同的齿轮定义不同的正方向。

Lossless Gears: reflecting impedance

- ▶ Substituting in for $\theta_2(s)$ gives

$$(Js^2 + Ds + K) \frac{N_1}{N_2} \theta_1(s) = T_1(s) \frac{N_2}{N_1}$$

- ▶ Simplifying: $[J(\frac{N_1}{N_2})^2 s^2 + D(\frac{N_1}{N_2})^2 s + K(\frac{N_1}{N_2})^2] \theta_1(s) = T_1(s)$

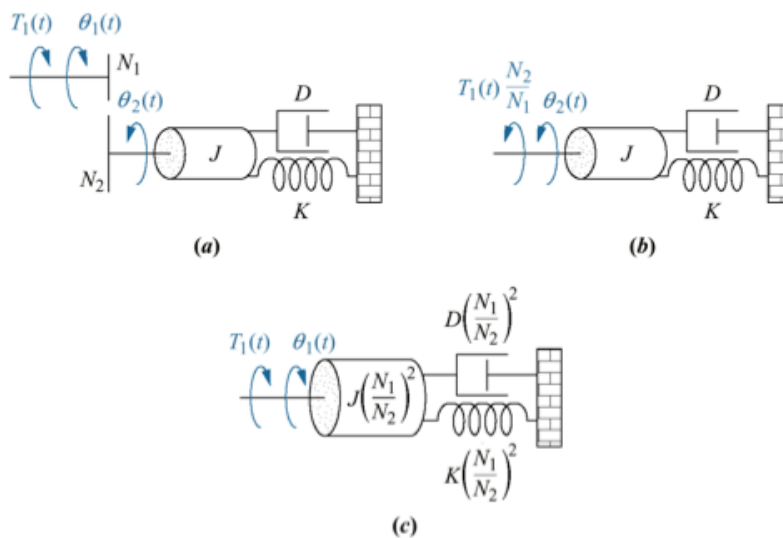


Figure 2.29.

Generalizing Reflecting Impedance

- ▶ **Rule:** "Rotational mechanical impedances can be reflected"

- Rule: Rotational mechanical impedances can be reflected through gear trains by multiplying the mechanical impedance by the ratio: “

$$\left(\frac{\text{Number of teeth of gear on destination shaft}}{\text{Number of teeth of gear on source shaft}} \right)^2$$

- ▶ The impedance to be reflected is attached to the source shaft and is being reflected to the destination shaft.
- ▶ In previous example gear 2 was the source shaft, and gear 1 was the destination shaft.

这里就涉及一个机械阻抗的化简问题，当两个转动系统通过齿轮相连时，可以把一侧轴上的机械阻抗“等效搬运”到另一侧轴上，但要乘一个齿轮比的平方，在这里就是把2的impedance等效到1上。

平方公式为： $\frac{N_{\text{destination}}^2}{N_{\text{source}}^2}$ ，source axis是原来impedance在的那个轴，destination axis是我想要等效到的那个轴。如果涉及到多级链接，先把远端的轴等效到中间轴，再等效到目标轴，一个一个来。

注意：等效只能允许刚性等效，即两个直接连接的刚性齿轮可以等效，因为等效的本质是 $r\theta$ 相等。我们做等效的本质是，一个刚性或者弹性物体，直接连接到一个刚性齿轮，然后我们可以等效，但是如果遇到那种弹性元件串联的轴，我们就不能随便把这个串联轴上的东西等效到其他轴上，一般来说就是把其他轴上的东西等效到他上面来，最后再通过齿轮比关系把我们想要的物理量比例乘回来。

Gear system做题建模步骤：

1. Gear system只有一个degree of freedom，因为齿轮之间只互相耦合的。我们先判断G是和哪个轴的旋转角度有关，然后我们直接把所有impedance都等效到这

个轴上。注意，我们对于齿轮有两个重要假设，即每两个相连接的齿轮在单位时间经过的arc length相等，即 $r\theta$ 相等；以及，两个齿轮的能量传递守恒，即 $T\theta$ 相等，由此我们推出 $\frac{\theta_2}{\theta_1} = \frac{N_1}{N_2} = \frac{r_1}{r_2} = \frac{T_1}{T_2}$ 。

2. 确定了G的组成之后，我们把其他轴的impedance都等效过来；注意，impedance的等效方式是乘以 $\frac{N_{\text{destination}}^2}{N_{\text{source}}^2}$ ，而torque的等效方式是通过 $N_1/N_2 = T_1/T_2$ 等效的，不要搞混。
3. 画出FBD，别忘了 $J s^2 \theta$ 的虚拟加速度力，列出一个微分方程（由于只有一个DOF），然后写出G。
4. 如果我们等效完发现系统是很多个DOF，那么不要慌，按照我们之前的做法做就行，如果发现我们G所需的物理量不在我们等效的轴上，那么先算出等效角度，然后我们按照齿轮比关系转换回去。

Lossless Gears eg.

- Find transfer function: $\frac{\theta_2(s)}{T_1(s)}$

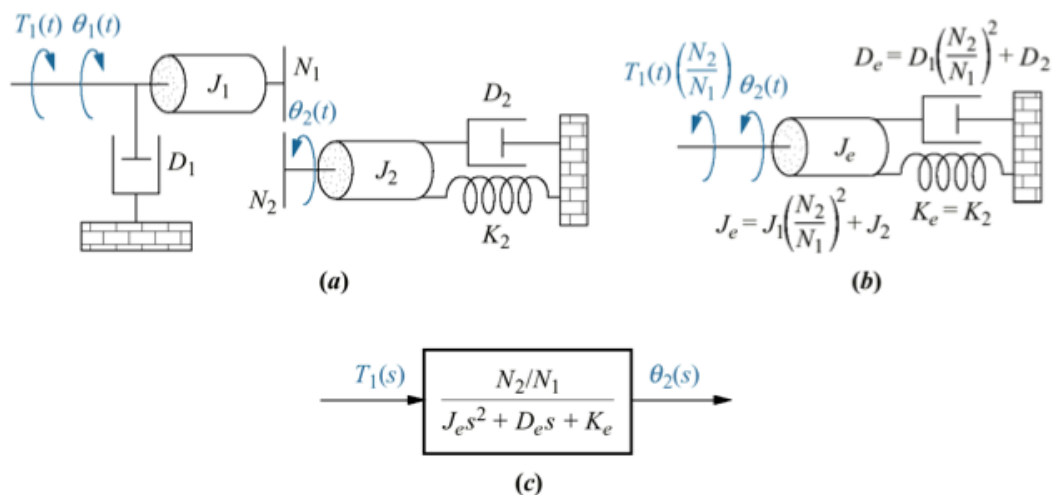


Figure 2.30.

$$\text{Find } G(s) = \frac{\theta_2(s)}{T_1(s)}$$

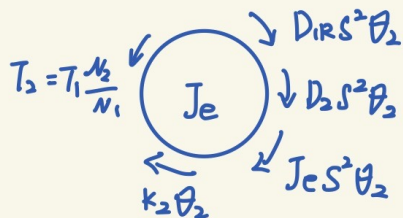
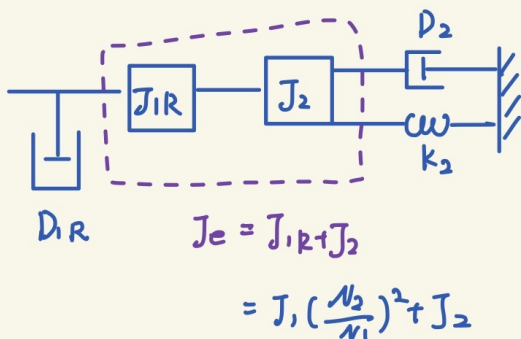
map onto θ_2 shaft

$$\textcircled{1} \text{ map } T_1(t) \rightarrow T_2 = \frac{N_2}{N_1} T_1$$

$\textcircled{2}$ Reflect impedances

destination is N_2 , source is N_1

$$J_{1R} = \left(\frac{N_2}{N_1}\right)^2 J_1, \quad D_{1R} = \left(\frac{N_2}{N_1}\right)^2 D_1$$



$$J_e s^2 \theta_2 + D_{1R} s \theta_2 + D_2 s \theta_2 + k_2 \theta_2 = T_1 \frac{N_2}{N_1}$$

$$(J_e s^2 + D_e s + k_e) \theta_2 = T_1 \frac{N_2}{N_1}$$

$$\frac{\theta_2}{T_1} = \frac{\frac{N_2}{N_1}}{J_e s^2 + D_e s + k_e}$$

$$= \frac{N_2/N_1}{J_e s^2 + D_e s + k_e}$$

$$(J_1(\frac{N_2}{N_1})^2 + J_2)\delta^2 + (D_1(\frac{N_2}{N_1})^2 + D_2)\delta + k_2$$

Gear Trains

- ▶ What if you wanted a high ratio, say $\frac{100}{1}$?
- ▶ To avoid a gear with $100\times$ as big radius, a **gear train** is used.
- ▶ The total equivalent gear ratio for the gear train, is the product of the gear ratios for each step.



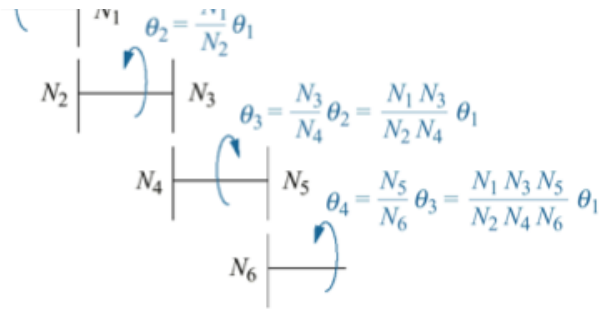


Figure 2.31.

For torques, we would have: $\frac{T_4}{T_1} = \frac{N_2 N_4 N_6}{N_1 N_3 N_5}$

Gears With Loss eg.

- Find transfer function $\frac{\theta_1(s)}{T_1(s)}$.
- Each gear below has its own inertia, and some shafts have nonnegligible damping.
- Need to reflect all moments of inertial and viscous dampers to the left of gear 1.

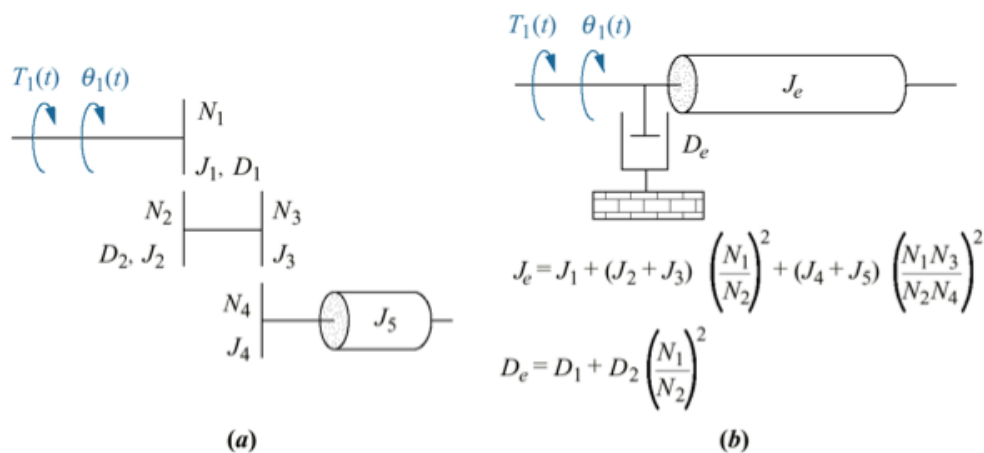


Figure 2.32.

同一个轴上的moment of inertia可以直接相加。考虑总damping和moment of inertia的时候记得不要忘记乘以齿轮比的平方。

Gears With Loss eg. - II

- Equation of motion:

$$(J_e s^2 + D_e s) \theta_1(s) = T_1(s)$$

- The transfer function is thus:

$$G(s) = \frac{\theta_1(s)}{T_1(s)} = \frac{1}{J_e s^2 + D_e s}$$

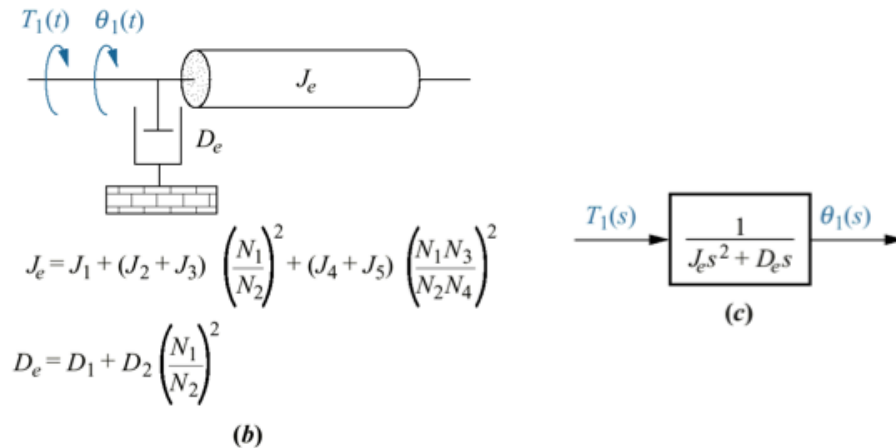


Figure 2.32.

Electromechanical System Transfer System

- **Electromechanical systems** are systems with a mixture of electrical and mechanical variables.

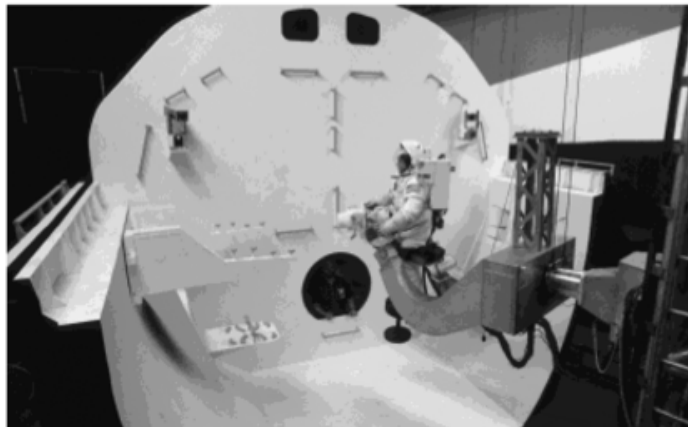


Figure 2.34. NASA flight simulator, Langley Research Center, VA

Figure 2.34: NASA flight simulator robot arm with electromechanical control system components.

DC Motors

- ▶ A motor takes a voltage as input and produces a physical displacement as output.
- ▶ We will derive the transfer function for the *armature-controlled dc servomotor*.

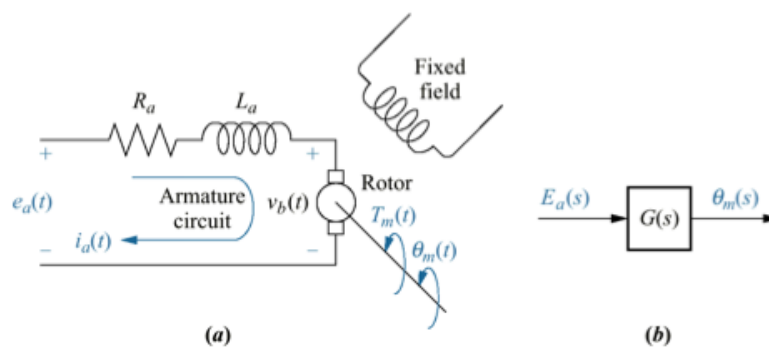
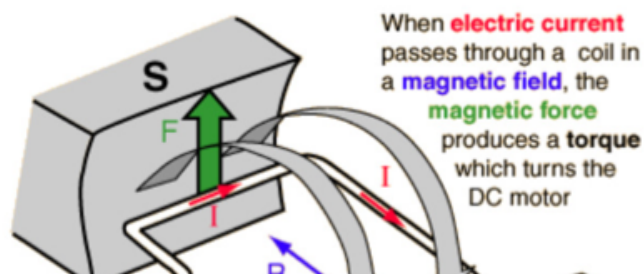
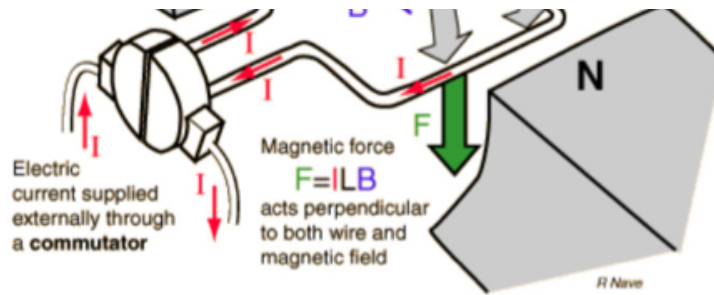


Figure 2.35.

DC Motor Principles





From: <http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/motdc.html>

DC Motors: basics

- ▶ DC motor contains stationary magnetic field provided by permanent or electromagnet with field strength B .
- ▶ Motor contains a rotating circuit called the **armature** (the “coils”) through which a current $i_a(t)$ flows.
- ▶ Current flows perpendicular through the magnetic field and feels a force $F = Bli_a(t)$ acting on it.
- ▶ Parameter l is the length of the conductor the current is flowing through within the field.
- ▶ Associated with the armature is a resistance R_a , and an inductance L_a .

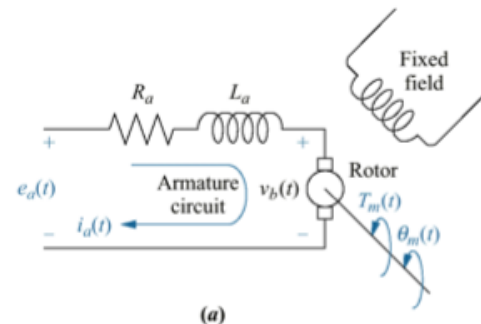


Figure 2.35.

DC motor建模步骤:

1. 先观察G是什么，找出我们想要的input/output，一般从G上找，确定我们要找的部分。
- 2 两个重要的关系式记住： $V_a(e) = K_v \omega(e)$ 和 $T(e) = K_t I(e)$ 。后面部分建模 生

2. 对1里写的关系式做拉氏变换： $V_b(s) = K_b s \theta_m(s)$ ，为内部搭建快，为建模电路部分，用KVL写出一个微分方程关系式，要把等式左右两边的变量（即Y(s)这种形式）化成G的分子分母，比如说G=Y(s)/X(s)，就把等式两边的变量都变成Y和X，。
3. 然后建模机械部分，得出关系式之后带入2的电路部分，得出一个只包含Y和X作为变量的微分方程，然后写出微分方程。

Back EMF

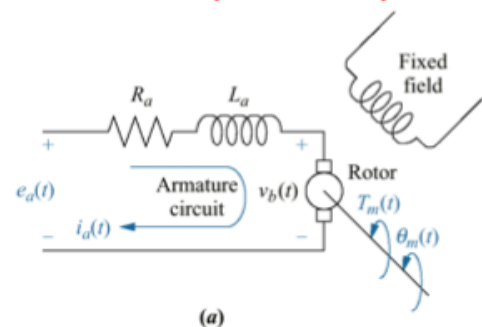
- ▶ When a conductor moves at right angles to a magnetic field, it generates a voltage at terminals of the conductor (think generator).
- ▶ Since armature is rotating in a magnetic field, it produces a voltage proportional to its velocity.

$$v_b(t) = K_b \frac{d\theta_m(t)}{dt}$$

- ▶ We refer to $v_b(t)$ as the **back electromotive force (back emf)**, and K_b as the *back emf constant*.

- ▶ Taking Laplace transform we get:

$$V_b(s) = K_b s \theta_m(s)$$



Analyzing Armature Circuit

- ▶ We wish to find the transfer function: $\frac{\theta_m(s)}{E_a(s)}$
- ▶ Applying Kirchhoff's voltage law to armature circuit gives:

$$R_a I_a(s) + L_a s I_a(s) + V_b(s) = E_a(s)$$

- ▶ The torque developed by motor is:

$$T_m(s) = K_t I_a(s) \Rightarrow I_a(s) = \frac{1}{K_t} T_m(s)$$

- ▶ Substituting in for $I_a(s)$ and $V_b(s)$ gives:

$$\frac{(R_a + L_a s) T_m(s)}{K_t} + K_b s \theta_m(s) = E_a(s)$$

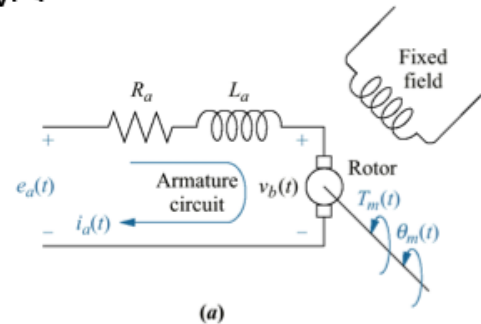


Figure 2.35.

Analyzing Armature Circuit - II

- ▶ Figure shows a typical mechanical loading of a motor.
- ▶ Variable J_m is the moment of inertia of armature plus that of the load reflected to the armature.
- ▶ Variable D_m is the viscous damping of the armature plus that of the load reflected to the armature.
- ▶ This gives the equation of motion:

$$T_m(s) = (J_m s^2 + D_m s) \theta_m(s)$$

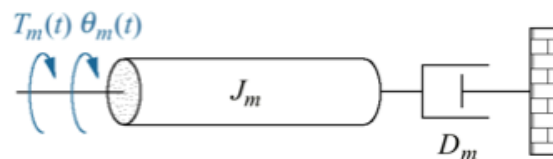


Figure 2.36.

DC Motor Transfer function

- ▶ We can now substitute into our previously derived equation for $E_a(s)$:

$$\frac{(R_a + L_a s)(J_m s^2 + D_m s)\theta_m(s)}{K_t} + K_b s\theta_m(s) = E_a(s)$$

- ▶ and noting that typically $(R_a + L_a s) \approx R_a$ gives:

$$\left[\frac{R_a}{K_t}(J_m s + D_m) + K_b \right] s\theta_m(s) = E_a(s)$$

- ▶ This gives the transfer function:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]} = \frac{K}{s(s + \alpha)}$$

Deriving J_m and D_m

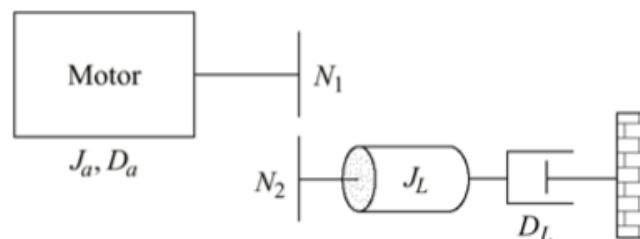


Figure 2.37.

- ▶ Figure shows typical usage of motor.
- ▶ Variable J_a is the moment of inertia of armature, and D_a is the viscous damping of the armature.
- ▶ Variable J_L is the load moment of inertia, and D_L is the viscous damping on the load.

- ▶ Reflecting onto the armature gives:

$$J_m = J_a + \left(\frac{N_1}{N_2}\right)^2 J_L \text{ and } D_m = D_a + \left(\frac{N_1}{N_2}\right)^2 D_L$$

Deriving Electrical Constants

- ▶ A *dynamometer* measures the torque and speed of a motor under a constant applied voltage.
- ▶ Previously, we derived:

$$\frac{R_a}{K_t} T_m(s) + K_b s \theta_m(s) = E_a(s)$$

- ▶ Inverse Laplace transform gives:

$$\frac{R_a}{K_t} T_m(t) + K_b \omega_m(t) = e_a(t)$$

- ▶ As voltage is constant, so is torque and velocity. Solving for T_m gives

$$T_m = -\frac{K_b K_t}{R_a} \omega_m + \frac{K_t}{R_a} e_a$$

Deriving Electrical Constants - II

$$T_m = -\frac{K_b K_t}{R_a} \omega_m + \frac{K_t}{R_a} e_a$$

- ▶ When $\omega_m = 0$ we get the *stall torque* given by:

$$T_{stall} = \frac{K_t}{R_a} e_a$$

- ▶ When $T_m = 0$ we get the angular velocity called *no-load*

speed given by: $\omega_{no-load} = \frac{e_a}{K_b}$

► We thus have:

$$\frac{K_t}{R_a} = \frac{T_{stall}}{e_a} \text{ and } K_b = \frac{e_a}{\omega_{no-load}}$$

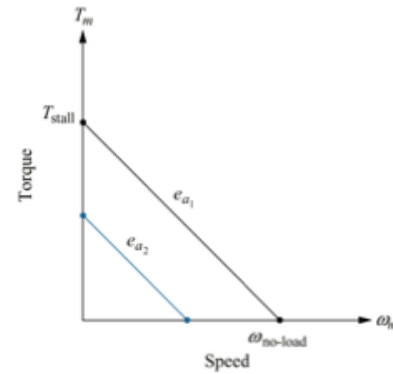


Figure 2.38.

对于这种把motor不告诉你细节的题，直接用这个transfer function：（或者只要L很小可以被忽略的题）

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_m}}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]} = \frac{K}{s(s + \alpha)}$$

然后通过上面的线性关系公式求出 $\frac{K_t}{R_a}$ 和 $\frac{K_t K_b}{R_a}$ 。注意： θ_m 是motor的角度，如果G需要的是别的轴的角度记得通过齿轮等效。

Linear Systems

- So far, we have developed models for systems that can be approximately represented by *linear, time-invariant differential equations*.
- In developing these models, we assumed they were **linear**.
- A **linear system** must have two properties:
 - Superposition:** means that the output response to a sum of inputs is equal to the sum of the output response of each individual input that makes up the “sum of inputs.”

ie. if input $r_1(t)$ generates output response $c_1(t)$, and input $r_2(t)$ generates output response $c_2(t)$, the the output response

to input $r_1(t) + r_2(t)$ will be $c_1(t) + c_2(t)$.

Linear Systems -II

Homogeneity: means that when the input is multiplied by a scalar, the result is a response multiplied by same scalar

ie. if input $r_1(t)$ generates output response $c_1(t)$, then input $Ar_1(t)$ will result in output $Ac_1(t)$. Below (a) is linear, (b) is not.

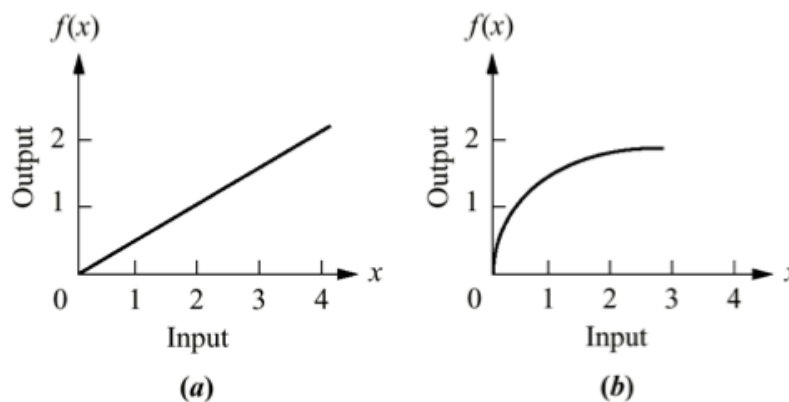


Figure 2.45.

Nonlinear Systems

- ▶ **Nonlinear systems** are systems that are not linear.
- ▶ Some examples are:
 - ▶ Op amps are linear over a given range, but exhibits **saturation** at high input voltages.
 - ▶ A motor exhibits a **deadzone** where it does not respond to low input voltages because of friction.
 - ▶ **Backlash** occurs when a pair of gears do not fit tightly. This is when the input gear moves through a small angle before the output gear starts to move.

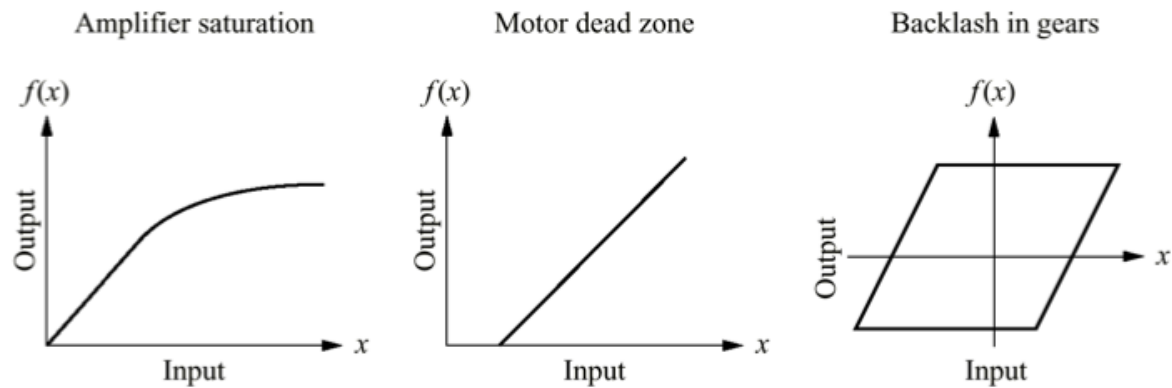


Figure 2.46.

Nonlinear Systems - II

- ▶ Sometimes we can make a linear approximation to a nonlinear system.
- ▶ For example, if system is linear for a portion, the range of input values about the point is small, and we can shift this portion to the origin, we can establish a linear relationship.
- ▶ Op amps are an example of this. We make sure our input values stay within the linear range.

Linearization

- ▶ If a system contains any nonlinear components, we must **linearize** the system before we can apply our linear systems

modelling methods.

- ▶ We need to find linear approximations in order to be able to find transfer functions.
- ▶ To linearize a system, we do the following:
 1. Identify the nonlinear components and write nonlinear differential equations for them.
 2. We then choose a small range of input values over which the system behaves approximately linear. Refer to this range as a *small-signal input*.

This range is centered around a steady state solution, called *equilibrium*, where the small-signal input is equal to zero.

3. We then write a linear differential expression for this range, and then apply Laplace transforms and form our transfer function as normal.

线性化微分方程的步骤:

1. 先确定G是什么, 和谁有关, 确定要找的物理量, 然后写出微分方程 (写出time domain的而不是s domain) 。
2. 线性化的核心在于找Taylor series展开的点, 我们叫做equilibrium point, 在该点处, 注意, 我们只对state variable找平衡点, state variable就是微分方程里面含有导数的变量, 把state variable写成 $x = x_0 + \delta x$ 的形式。平衡点的条件是导数为0, 同时输入为小信号 (根据题目的要求带入数值), 求出此时 x_0 的数值。
3. 找到equilibrium point之后, 然后把x带入回微分方程, 找其不是线性函数的部分做线性化, 用Taylor series的一阶展开来近似, 即 $f(x) - f(x_0) = f'(x_0) * \delta x$, 求出f(x)的表达式, 注意不要忘了消去微分方程里面的常数项(如果给input和output都写成了 $x = x_0 + \delta x$ 的形式), 因为我们只关心小扰动的动态关系。
4. 用Laplace domain来表示微分方程并且求出小信号输入的表达式之后, 找出G所需的另一个物理量, 用Laplace和物理公式来表示他, 求出G。

Linearizing a Function

- ▶ Assume we have a nonlinear system that is operating about

point A: $[x_o, f(x_o)]$.

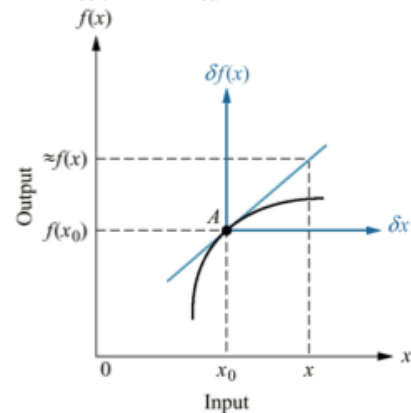
- ▶ For small changes around A, we can calculate the approximate value of $f(x)$ using the slope of the curve at x_o .
- ▶ If slope at point A is m_a we can determine the change in $f(x)$, called $\delta f(x)$, for very small changes in x , called δx .

$$\delta f(x) = [f(x) - f(x_o)] \approx m_a(x - x_o) = m_a \delta x$$

- ▶ This gives us:

$$\begin{aligned} f(x) &\approx f(x_o) + m_a(x - x_o) \\ &\approx f(x_o) + m_a \delta x \end{aligned}$$

Figure 2.47.



Linearizing a Function - eg.

- ▶ Linearize $f(x) = 5 \cos x$ about $\pi/2$.

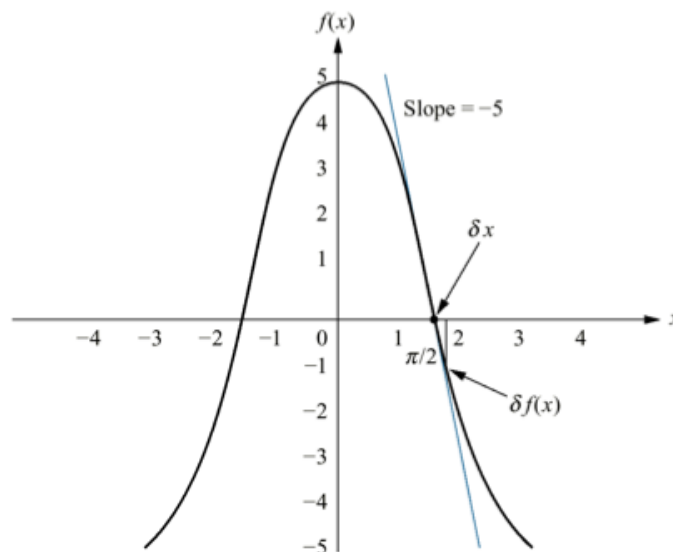


Figure 2.48.

To get slope of eqn at $x_o = \pi/2$ we take the derivative of $f(x)$:

$$f'(x) = \frac{df}{dx}(x) = \frac{d}{dx}(5 \cos x) = -5 \sin x$$

Linearizing a Function - eg. - II

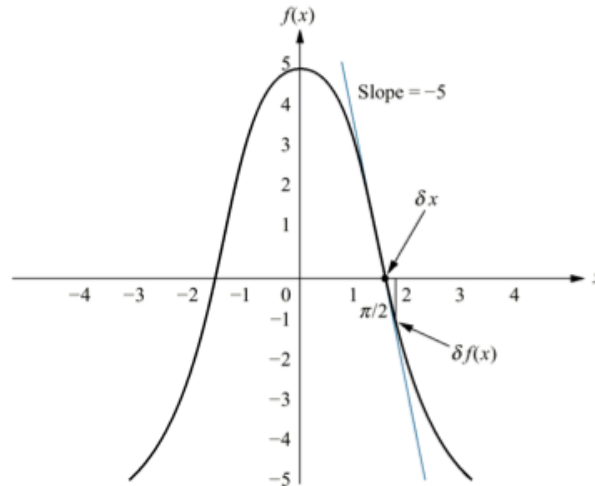


Figure 2.48.

Thus: $f'(\pi/2) = -5 \sin(\pi/2) = -5 = m_a$

Also: $f(x_o) = f(\pi/2) = 5 \cos(\pi/2) = 0$

We thus have:

$$f(x) = f(x_o) + m_a \delta x = 0 - 5\delta x = -5\delta x = -5(x - x_o)$$

Formalizing the Linearization Method.

- ▶ Using the **Taylor series expansion**, we can express the value of a function about point x_o using the equation:

$$f(x) = f(x_o) + \left. \frac{df}{dx} \right|_{x=x_o} \frac{(x - x_o)}{1!} + \left. \frac{d^2 f}{dx^2} \right|_{x=x_o} \frac{(x - x_o)^2}{2!} + \dots$$

- ▶ For small range around x_o , we can neglect higher order terms and we get:

$$f(x) - f(x_o) \approx \left. \frac{df}{dx} \right|_{x=x_o} (x - x_o) = m|_{x=x_o} (x - x_o)$$

Linearizing a Differential Eqn eg.

- Linearize the differential equation below about $x_o = \pi/4$:

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + \cos x = 0$$

First, rewrite equation in terms of δx . As $\delta x = x - x_o$, we get $x = \delta x + x_o = \delta x + \pi/4$ giving:

$$\frac{d^2(\delta x + \pi/4)}{dt^2} + 2\frac{d(\delta x + \pi/4)}{dt} + \cos(\delta x + \pi/4) = 0$$

As the derivative of a constant is zero, the equation simplifies to:

$$\frac{d^2\delta x}{dt^2} + 2\frac{d\delta x}{dt} + \cos(\delta x + \pi/4) = 0 \quad (7)$$

We now need to evaluate: $f(x) - f(x_o) \approx \left. \frac{df}{dx} \right|_{x=x_o} \delta x$

最后一行来自于Taloy series, 但是只保留一阶项。

Linearizing a Differential Eqn eg. - II

We now need to evaluate:

$$f(x) - f(x_o) \approx \left. \frac{df}{dx} \right|_{x=x_o} \delta x \quad (8)$$

Our function is: $f(x) = \cos(x) = \cos(\delta x + \pi/4)$

We have $f(x_o) = f(\pi/4) = \cos(\pi/4)$ and

$$\left. \frac{d(\cos x)}{dx} \right|_{x=\pi/4} = -\sin(\pi/4)$$

Subbing into 8 gives:

$$\cos(\delta x + \pi/4) - \cos(\pi/4) = -\sin(\pi/4)\delta x$$

Linearizing a Differential Eqn eg. - III

Solving for $f(x)$ gives:

$$\cos(\delta x + \pi/4) = \cos(\pi/4) - \sin(\pi/4)\delta x = 0.7071 - 0.7071\delta x$$

Subbing into 7 gives:

$$\frac{d^2 \delta x}{dt^2} + 2 \frac{d\delta x}{dt} - 0.7071\delta x = -0.7071$$

We now have a linear differential equation.

If we wished to solve for x , we could take the Laplace transform of both sides (taking δx as a function of x) assuming zero initial conditions, solve for $\delta x(s)$, do inverse Laplace transform, and then sub in for $\delta x = x - x_o$.

Transfer Function - Nonlinear Electrical System eg.

- Find transfer function $\frac{V_L(s)}{V(s)}$, where voltage current relationship for resistor is $i_r = 2e^{0.1v_r}$, and $v(t)$ is a small-signal voltage source.

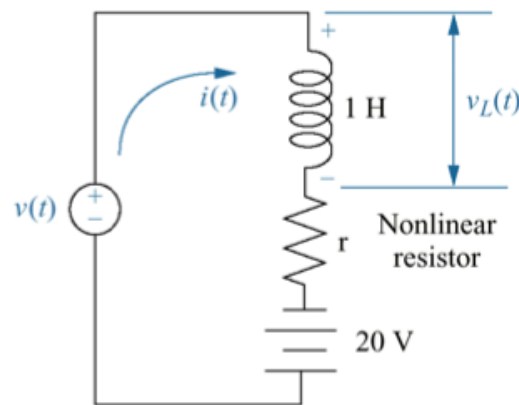


Figure 2.49.

$$G(s) = \frac{V_L(s)}{V(s)}$$

① Find DE:

$$\because i_r = 2e^{0.1V_r} \rightarrow \frac{i_r}{2} = e^{0.1V_r} \quad \therefore 0.1V_r = \ln \frac{i_r}{2} \rightarrow V_r = 10 \ln \frac{i_r}{2}$$

$$\therefore \text{KVL: } V(t) - 10 \ln \frac{i}{2} - 2 \frac{di}{dt} + 20 = 0$$

② Find equilibrium point:

$$\text{let } i' = 0, \quad V(t) = 0 \quad \text{let } i = i_0 + \delta i$$

$$\therefore -10 \ln \frac{i}{2} + 20 = 0$$

$$\therefore i = 2e^{-2} + \delta i$$

$$\ln \frac{i}{2} = 2$$

$$\frac{i}{2} = e^2$$

$$i = 2e^2$$

$$\textcircled{3} \text{ Linearize } -10 \ln \frac{i}{2}, \quad (-10 \ln \frac{i}{2})' = -10 \cdot \frac{1}{\frac{i}{2}} \cdot \frac{1}{2} = -\frac{10}{i}$$

$$\therefore -10 \ln \frac{i}{2} + 10 \ln \frac{i_0}{2} = -\frac{10}{i_0} \cdot \delta i$$

$$\therefore -10 \ln \frac{i}{2} = -10 \ln \frac{2e^2}{2} - \frac{10}{2e^2} \cdot \delta i = -20 - \frac{5\delta i}{e^2}$$

$$\therefore \mathcal{L} \frac{d\delta i}{dt} + 20 + \frac{5\delta i}{e^2} - 20 = V(s)$$

$$\therefore \mathcal{L} s I(s) + \frac{5}{e^2} I(s) = V(s)$$

$$\therefore I(s) = \frac{V(s)}{s + \frac{5}{e^2}}$$

$$\therefore V_L(s) = I(s) \cdot sL = \frac{sV(s)}{s + \frac{5}{e^2}} = \frac{sV(s)}{s + 0.68}$$

$$\therefore G(s) = \frac{V_L(s)}{V(s)} = \frac{s}{s + 0.68}$$

这道题没给V找equilibrium point是因为这里面已经定义了输入V是small signal，平衡点就是0。